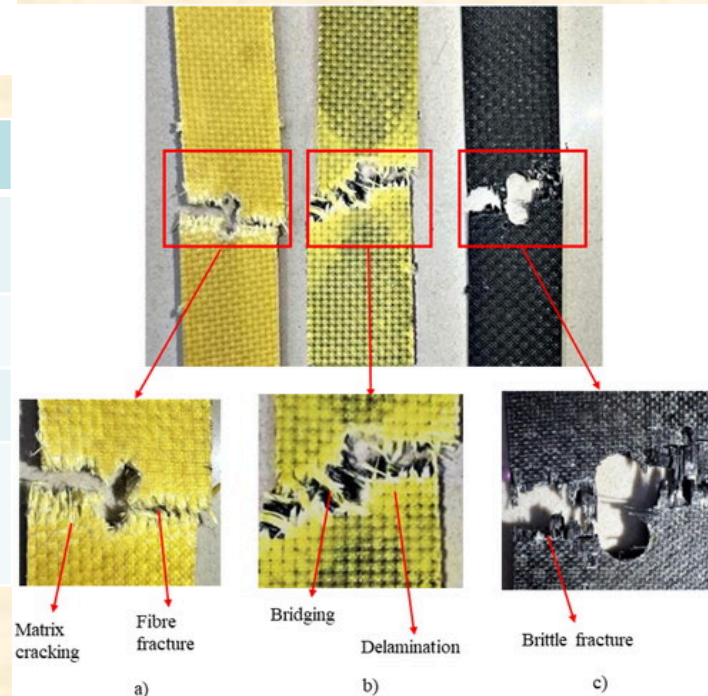
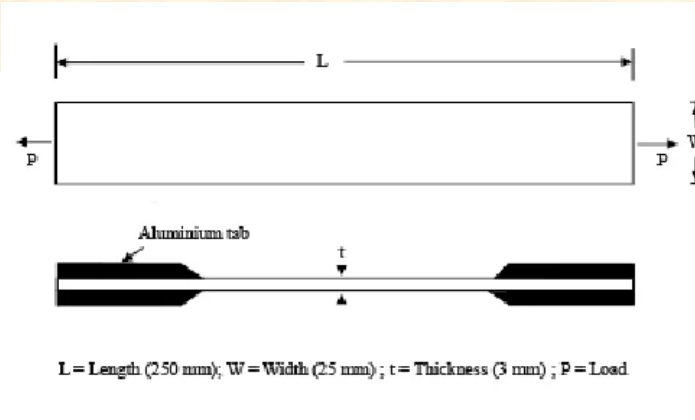


Tensile testing of Composites

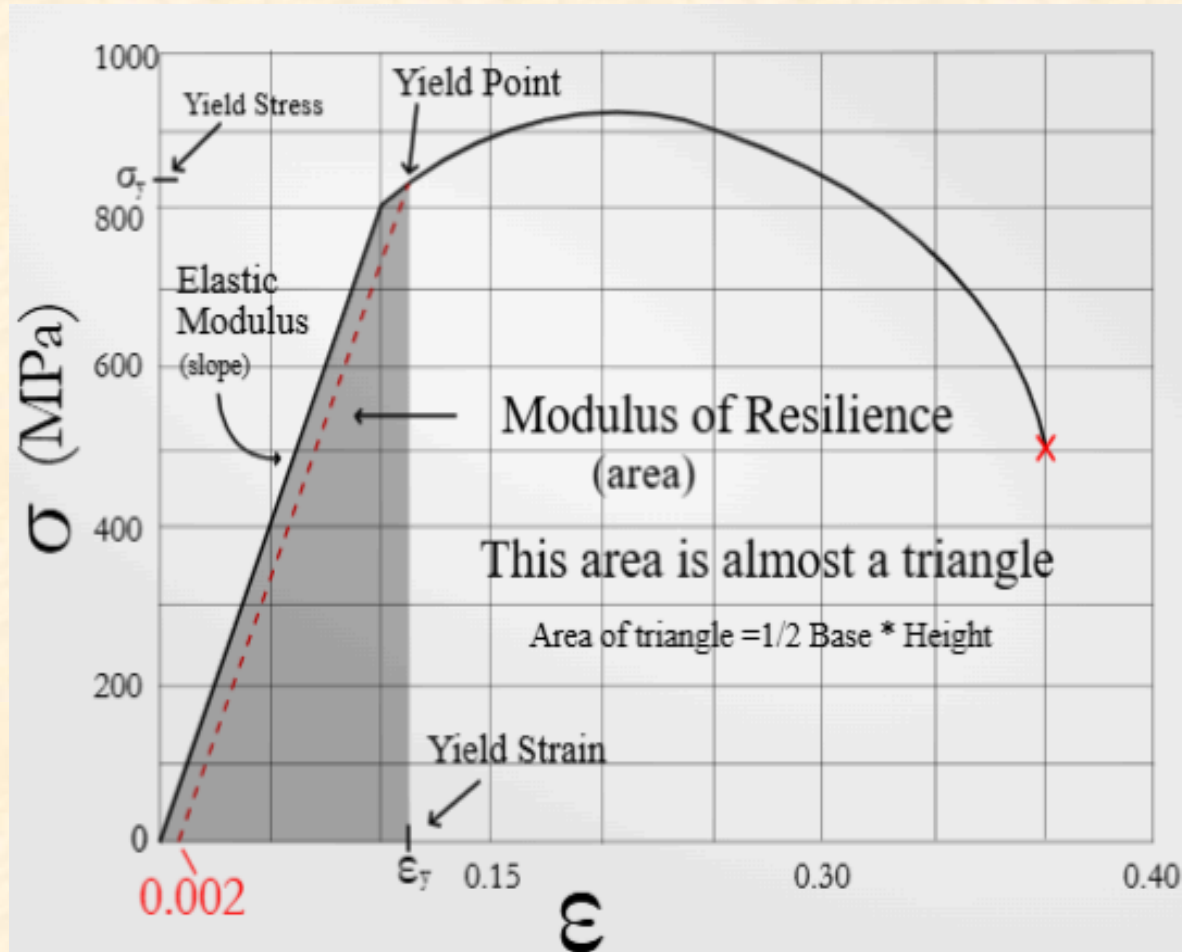
Several variables influence the final tensile performance of a composite material:

- Fiber type (carbon, glass, aramid, basalt, or natural fibers)
- Fiber volume fraction
- Resin formulation
- Laminate stacking sequence
- Fiber orientation
- Curing temperature and pressure
- Internal void content
- Fiber–matrix interfacial bonding
- Moisture absorption and environmental aging



Resilience

Resilience is a material's ability to absorb energy and resist shock and impact loads. Resilience is the total strain energy stored in a given volume of material within the elastic limit, and it's released when the load is removed.



Resilience

Elastic energy that can be stored and released

Modulus of resilience is like toughness, but only up to the yield strain.

$$U_r \approx \frac{1}{2} \epsilon_y \sigma_y$$

Resilient materials have high yield strain and low elastic modulus

Area under the whole tensile curve upto fracture is known as Toughness

Resilient Substances

Natural rubber

0 1

0 2

Synthetic polymers

Metals

0 3

0 4

Composite materials

Biological materials

0 5





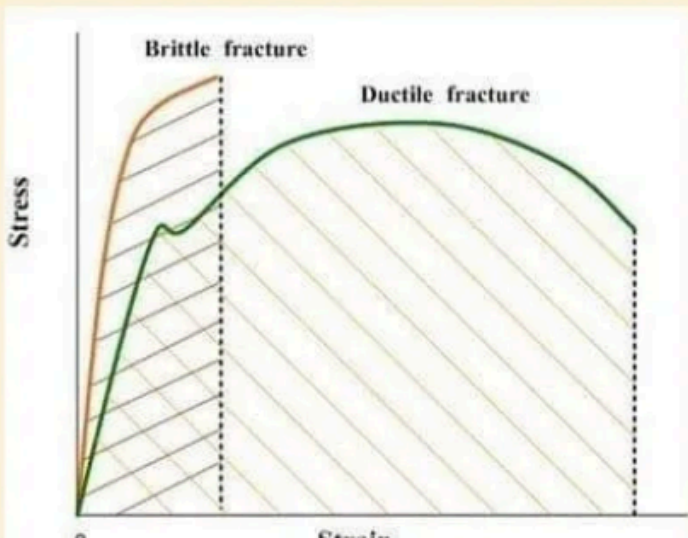
Toughness vs Resilience



Insta- metallurgical_engineering

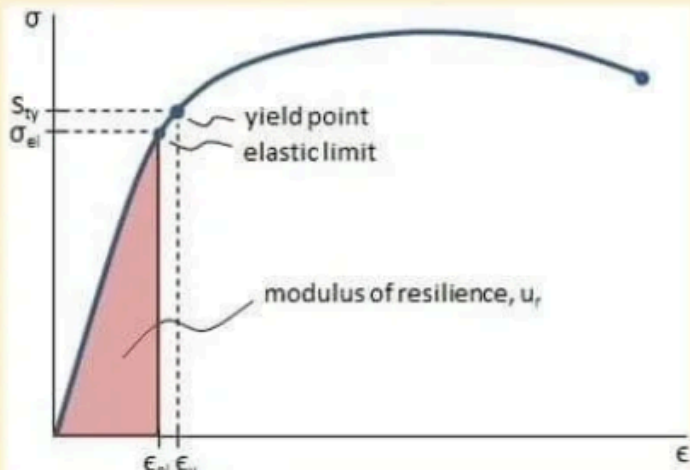
Toughness

- The *modulus of toughness* is the amount of strain energy per unit volume (i.e. strain energy density) that a material can absorb just before it fractures.
- The modulus of toughness is calculated as the area under the stress-strain curve up to the fracture point.
- An accurate calculation of the total area under the stress-strain curve to determine the modulus of toughness is somewhat involved.



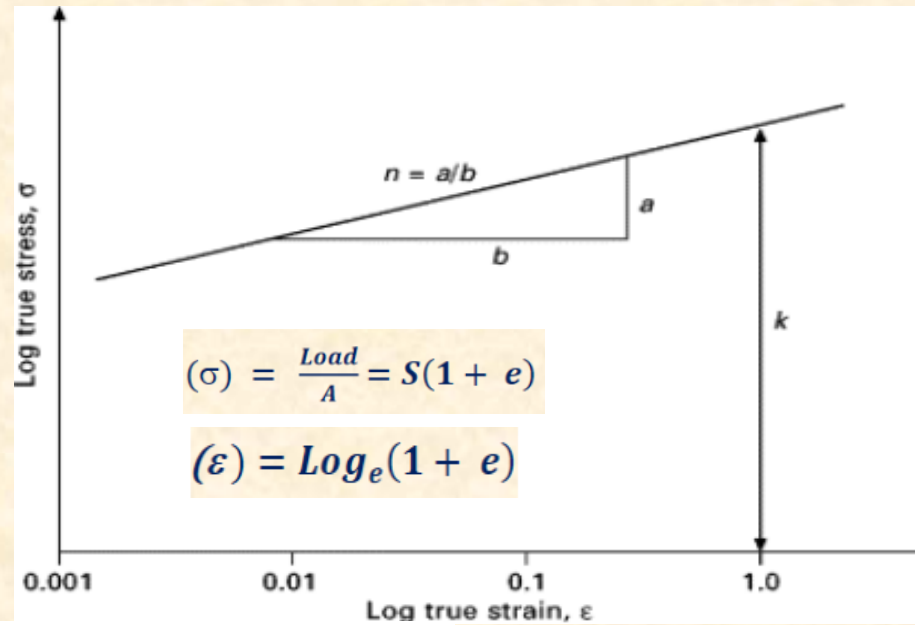
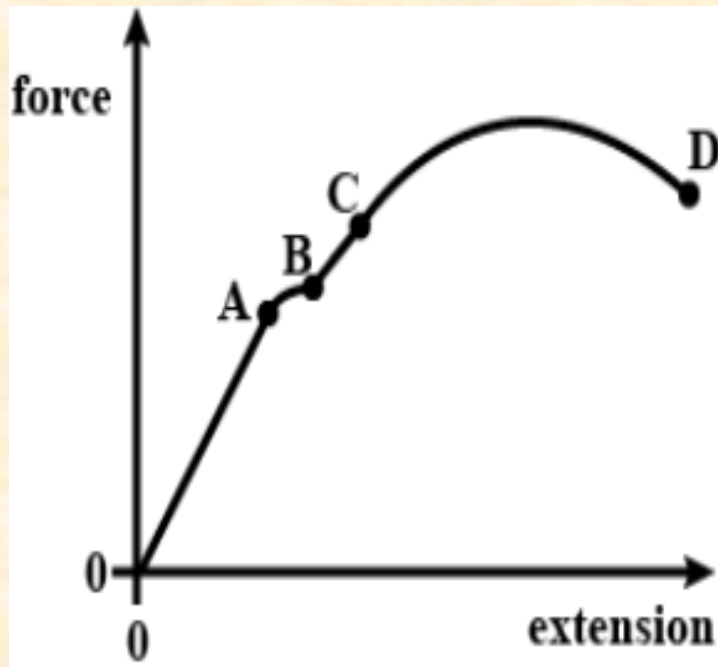
Resilience

- The *modulus of resilience* is the amount of strain energy per unit volume (i.e. strain energy density) that a material can absorb without permanent deformation resulting.
- The modulus of resilience is calculated as the area under the stress-strain curve up to the elastic limit. However, since the elastic limit and the yield point are typically very close, the resilience can be approximated as the area under the stress-strain curve up to the yield point.



Strain Hardening

Strain Hardening exponent, n



<i>Material</i>	<i>n</i>	<i>K in MPa</i>
Low-carbon steel (annealed)	0.21	600
4340 steel alloy (tempered @ 315°C)	0.12	2650
304 stainless steel (annealed)	0.44	1400
Copper (annealed)	0.44	530
Naval brass (annealed)	0.21	585
2024 aluminum alloy (heat treated—T3)	0.17	780
AZ-31B magnesium alloy (annealed)	0.16	450

$$\sigma = K \epsilon^n$$

K : Strain Hardening Coefficient

n : Strain Hardening exponent

Determination of Strain Hardening (alternate method)

In a tensile test, a metal begins to neck at a true strain of $\epsilon_t = 0.36$ with a corresponding true stress of $\sigma_t = 550.0$ MPa. Find the strain-hardening exponent (n) and the strength coefficient (K).

For a metal undergoing tensile deformation, the onset of necking corresponds to the condition where true strain equals the strain hardening exponent

Identify the governing equation: The plastic deformation follows the Hollomon flow curve equation:

$$\sigma_t = K \cdot \epsilon_t^n$$

Determine the strain-hardening exponent (n): At the onset of necking (instability point), the true strain equals n :

$$n = 0.36$$

Substitute values into the equation to find K :

$$550.0 \text{ MPa} = K \cdot (0.36)^{0.36}$$

Calculate K :

$$(0.36)^{0.36} \approx 0.692 \quad K = \frac{550.0 \text{ MPa}}{0.692} \approx 794.5 \text{ MPa}$$

Question

During a uniaxial tensile test, a metal specimen demonstrates a **true stress of 250 MPa at a true strain of 0.05**. When the **true strain increases to 0.15**, the true stress rises to **380 MPa**. Assuming the material obeys Hollomon's equation ($\sigma = K\epsilon^n$), calculate the **strain hardening exponent (n)** and the **strength coefficient (K)**

Solution

1. **Set up the simultaneous equations** using the raw data points:

$$\sigma_1 = K\epsilon_1^n \implies 250 = K(0.05)^n \quad \text{Eqn 1} \qquad \sigma_2 = K\epsilon_2^n \implies 380 = K(0.15)^n \quad \text{Eqn 2}$$

2. **Isolate the strain hardening exponent (n)** by dividing the two equations:

$$\frac{380}{250} = \left(\frac{0.15}{0.05}\right)^n \implies 1.52 = (3)^n$$

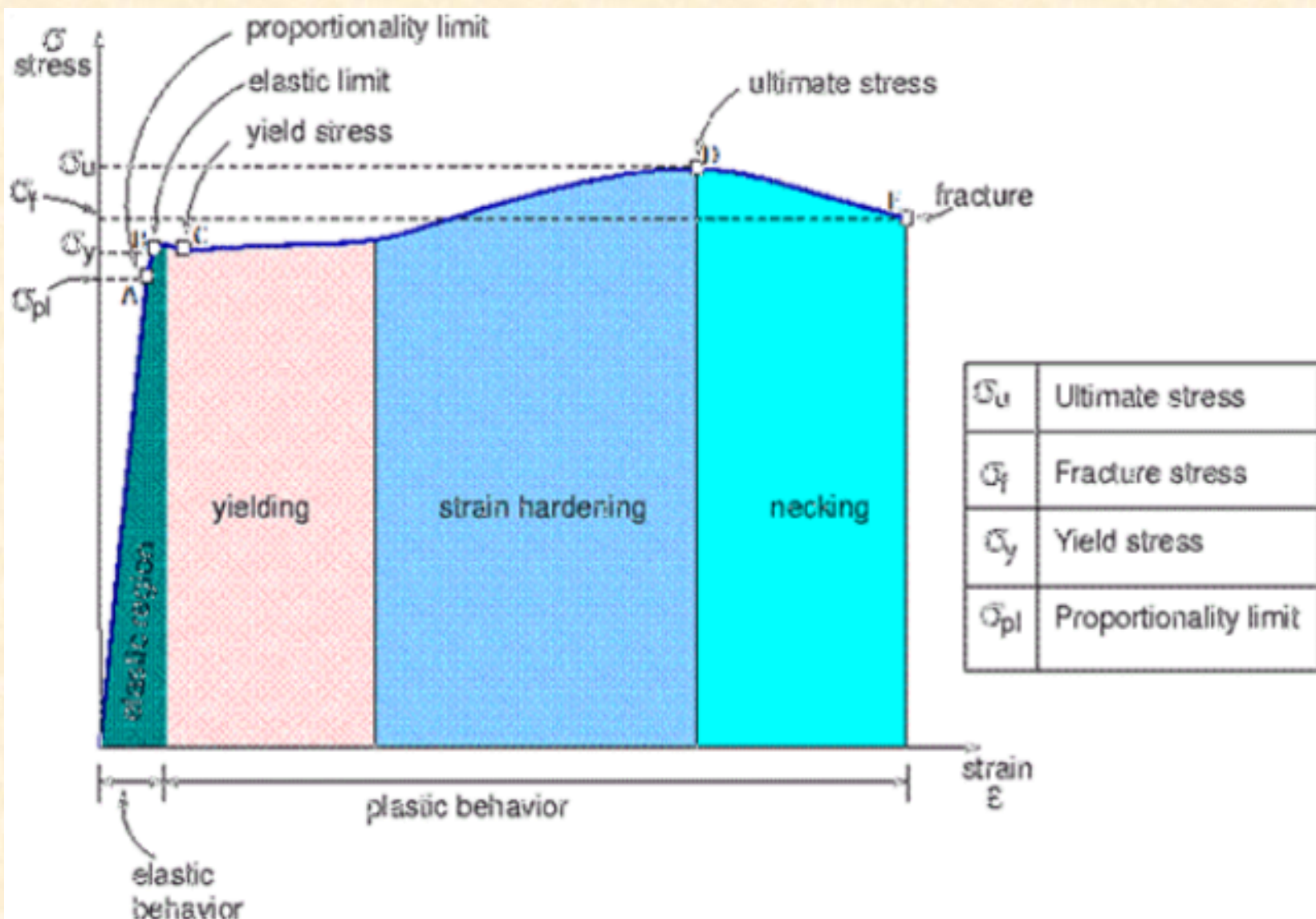
3. **Take the natural logarithm ($\ln$)** of both sides to solve for n :

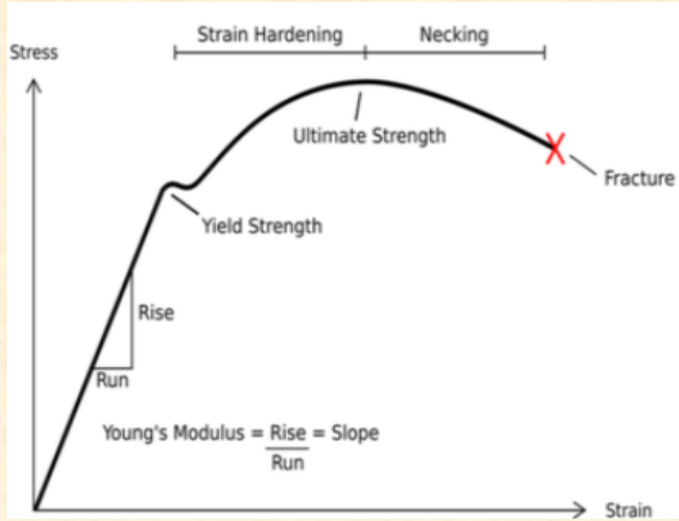
$$\begin{aligned} \ln(1.52) &= n \cdot \ln(3) \\ 0.4187 &= n \cdot 1.0986 \implies \mathbf{n \approx 0.381} \end{aligned}$$

4. **Substitute n back** into the first equation to isolate the strength coefficient (K):

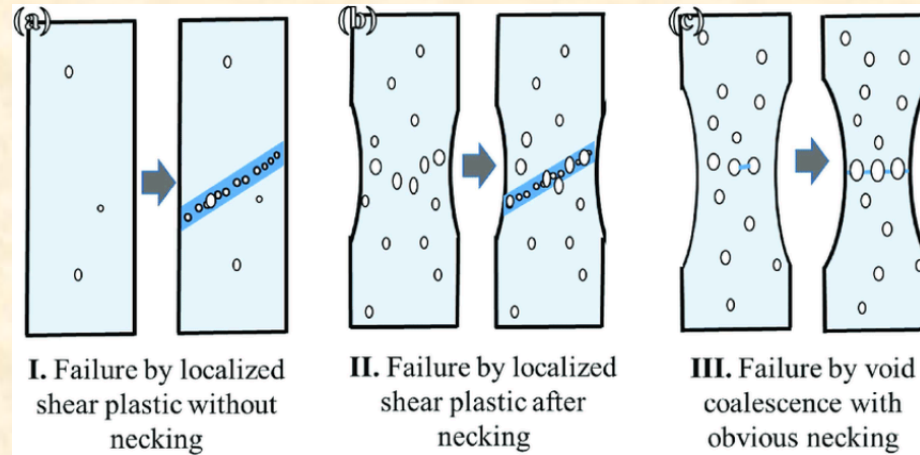
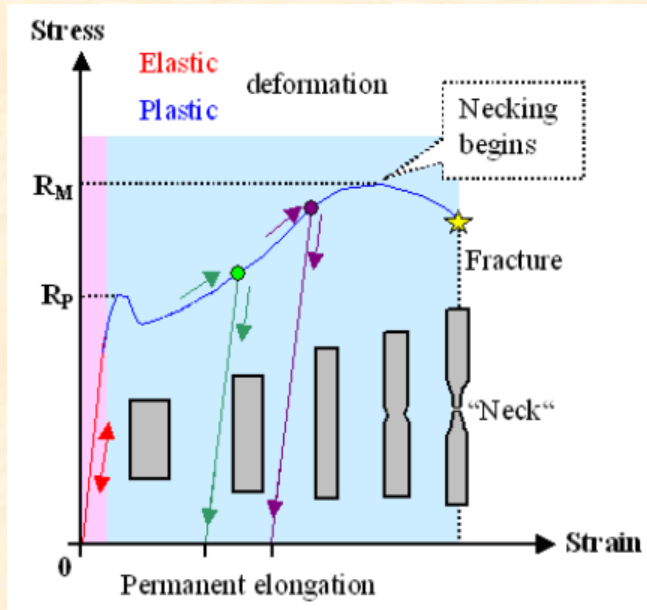
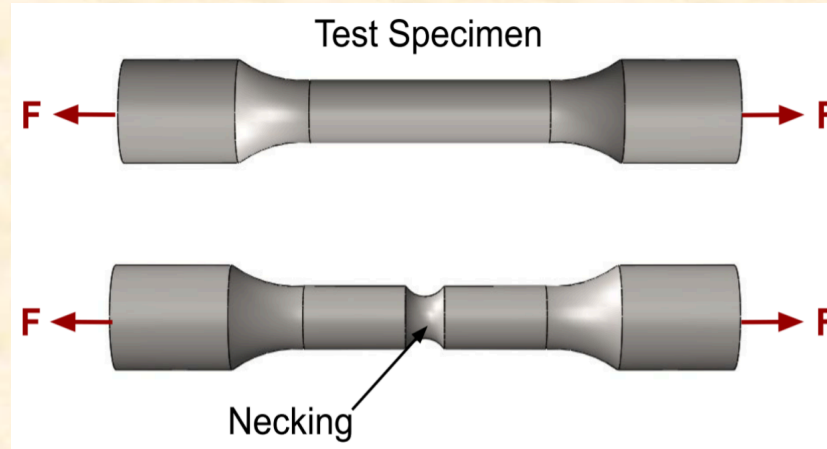
$$\begin{aligned} 250 &= K(0.05)^{0.381} \\ 250 &= K(0.3204) \implies K = \frac{250}{0.3204} \implies \mathbf{K \approx 780.3 \text{ MPa}} \end{aligned}$$

Necking and Strain Rate Sensitivity





Necking



Strain Rate sensitivity

A series of tensile tests are performed at an elevated temperature to study rate-dependent effects. At a constant true plastic strain of 0.20, a strain-rate jump is performed. The **true stress is 120 MPa** when deforming at a **strain rate of 10^{-3} s^{-1}** . When the **strain rate is abruptly increased to 10^{-1} s^{-1}** , the flow stress jumps to **165 MPa**. Find the **strain rate sensitivity index (m)**.

Solution

$$\sigma_t = C \dot{\epsilon}_t^m$$

$$m = \frac{\ln(\sigma_2) - \ln(\sigma_1)}{\ln(\dot{\epsilon}_2) - \ln(\dot{\epsilon}_1)}$$

1. **Apply the strain rate sensitivity formula** relating two distinct data points:

$$m = \frac{\ln(\sigma_2/\sigma_1)}{\ln(\dot{\epsilon}_2/\dot{\epsilon}_1)}$$

2. **Substitute the given metrics** into the logarithmic ratio:

$$m = \frac{\ln(165/120)}{\ln(10^{-1}/10^{-3})} \quad m = \frac{\ln(1.375)}{\ln(100)} \quad m = \frac{0.3185}{4.6052} \Rightarrow \mathbf{m \approx 0.069}$$

$$\sigma_t = C \dot{\epsilon}_t^m$$

Strain Rate sensitivity

Step-by-Step Testing and Calculation Method

1. Perform Mechanical Tests at Multiple Speeds

Conduct uniaxial tensile tests at least at two (preferably more) different strain rates ($\dot{\epsilon}_1, \dot{\epsilon}_2, \dot{\epsilon}_3$), ranging from slow quasi-static speeds to dynamic rates.

Keep the test temperature constant

Record the resulting stress-strain curves and extract the flow stress at a fixed plastic strain for each rate.

$$m = \frac{\ln(\sigma_2) - \ln(\sigma_1)}{\ln(\dot{\epsilon}_2) - \ln(\dot{\epsilon}_1)}$$

2. Take the Logarithm of the Equation

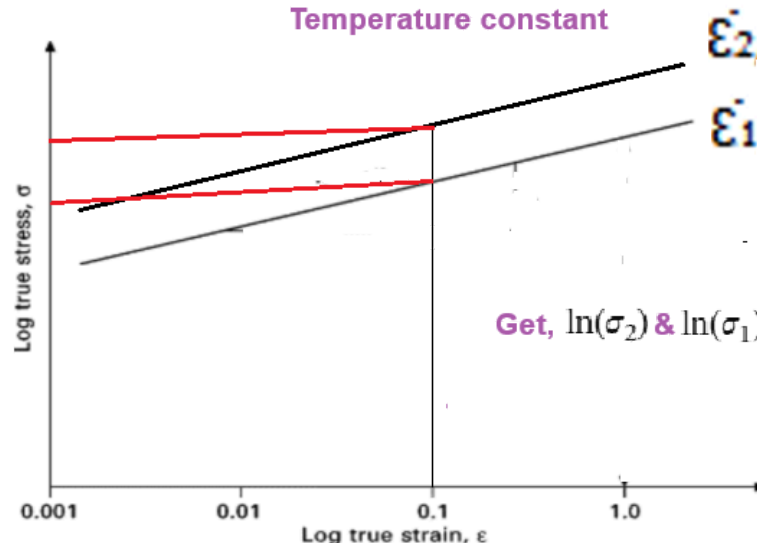
Take the natural logarithm (ln) or base-10 logarithm (log) of the power-law equation:

$$\ln(\sigma) = m \ln(\dot{\epsilon}) + \ln(C)$$

Role in Superplasticity

A material is classified as superplastic if m value is more than 0.3. Ideal value is around 0.7

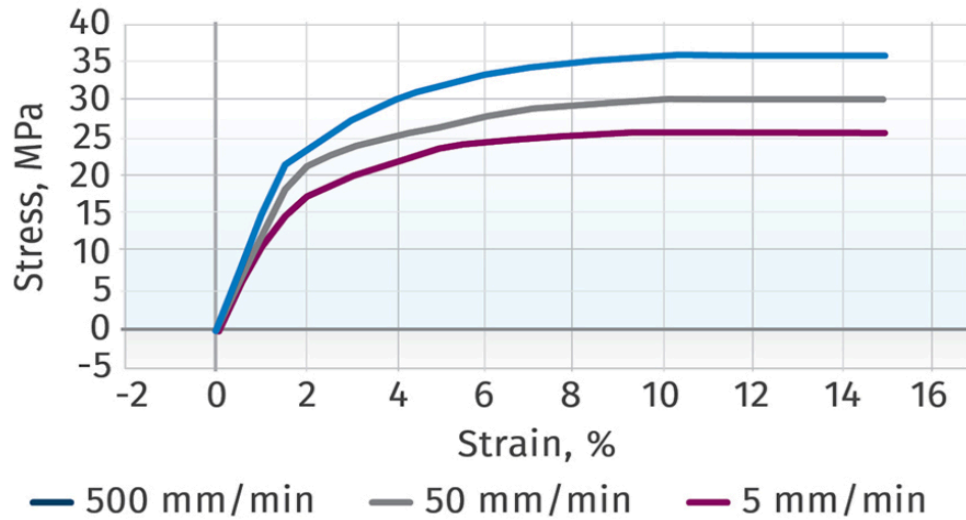
These materials have high necking resistance and prevents premature tearing. and allows plastic deformation to reach as high as 2000%



$$\sigma_t = C \dot{\epsilon}_t^m$$

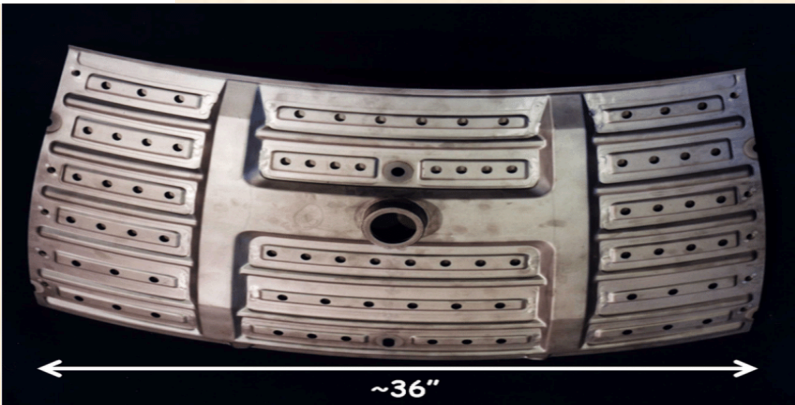
Strain Rate sensitivity

Effect of Strain Rate on Stress-Strain Properties of a Polypropylene



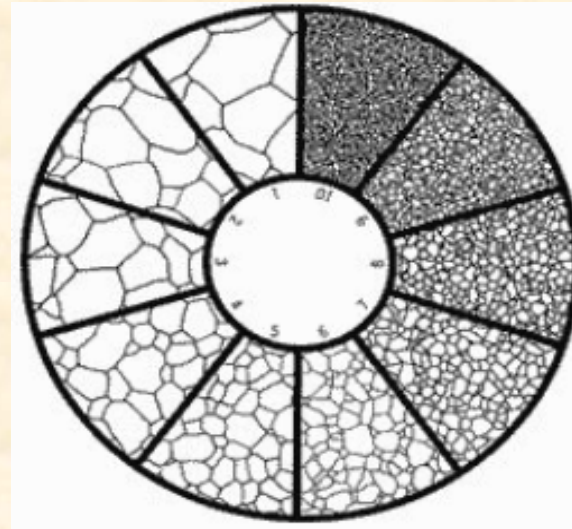
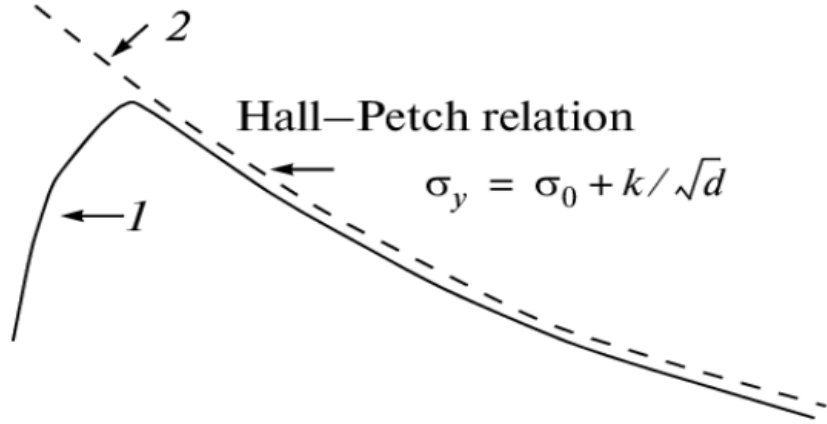
Production of ultrafine grain size Strain rate sensitivity index "m"

1. m increases with decreasing grain size
2. m increases with increasing temperature
3. In many alloy system
maximum value of 'm' is reached at a temperature just below the phase boundary defining the upper limit of the two phase field .



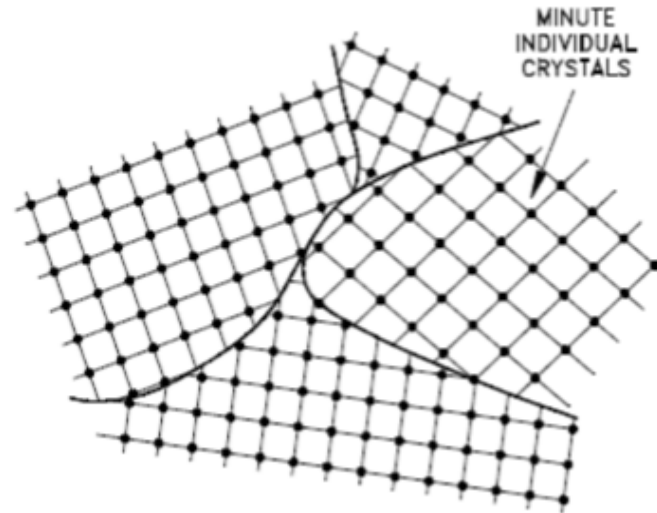
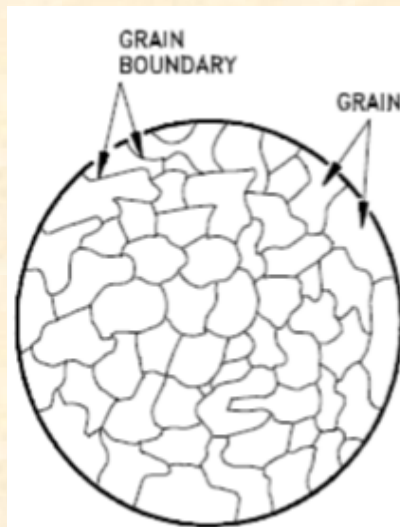
Factors that influence Tensile strength

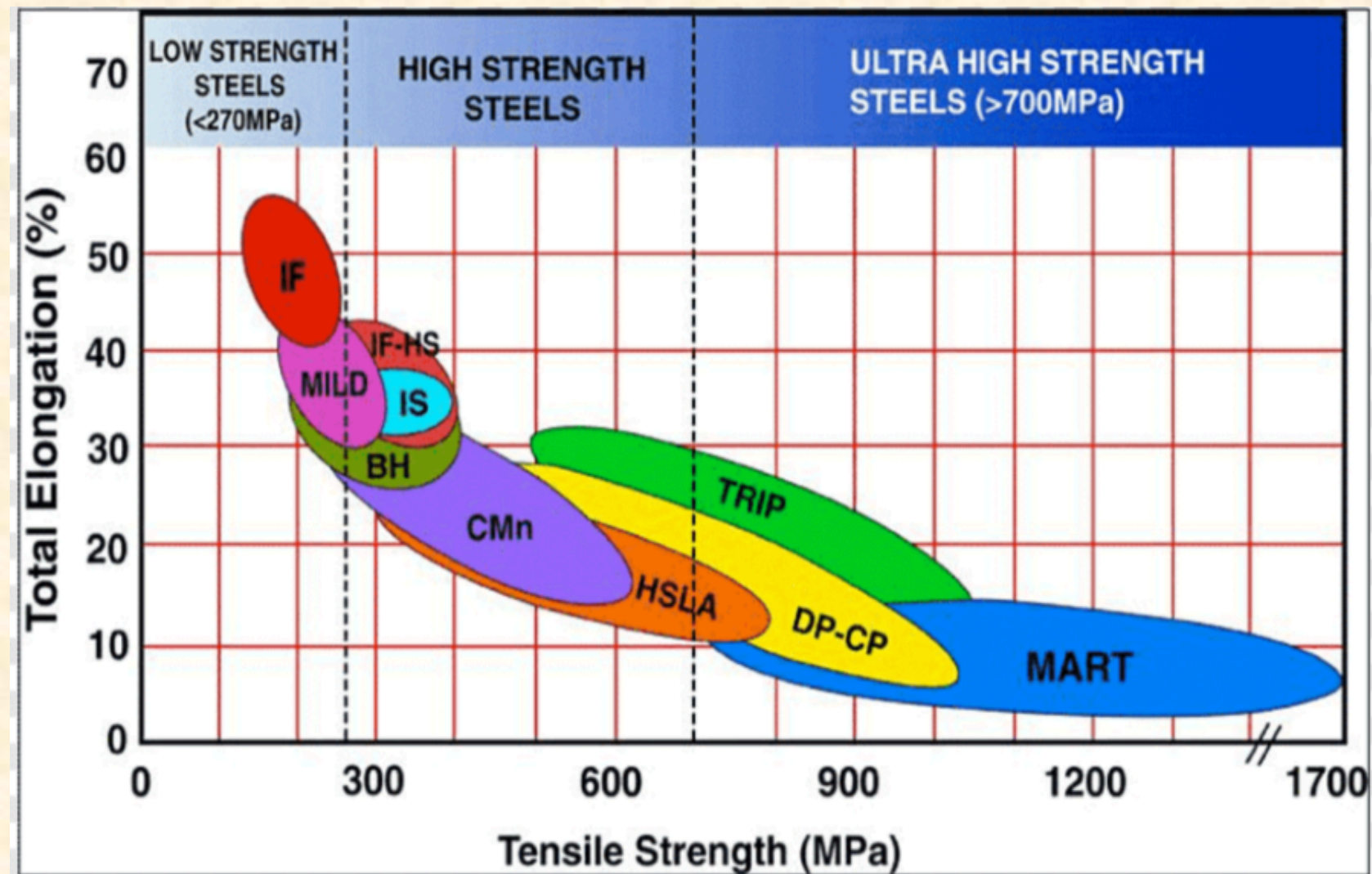
Theoretical yield strength σ_t



Grains act as impediment to moving dislocations and aid in improving Tensile strength

ASTM Grain size varies from 0 to 15. A size 0 indicates coarse grain







Material Technology & Testing (DE ZG631)

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Introduction to Fracture Mechanics



Fracture Mechanics

Introduction

Why Structures Fail

- ❑ Negligence during design, construction, or operation of the structure ----- **Type- 1**
- ❑ Application of a new design or material, which produces an unexpected (and undesirable) result ----- **Type- 2**



Fracture Mechanics

Introduction

- ❑ Negligence during design, construction, or operation of the structure ----- **Type- 1**

due to human error, ignorance, or willful misconduct. Poor workmanship, inappropriate or substandard materials, errors in stress analysis, and operator error

Example: Boston Molasses Tank Failure



Fracture Mechanics

Introduction

- ❑ Application of a new design or material, which produces an unexpected (and undesirable) result ----- **Type- 2**

Improved design is introduced however there are factors that the designer does not anticipate.

New materials can offer tremendous advantages, but also potential problems.

Example: Liberty Ship Failure, Comet Disaster



Introduction

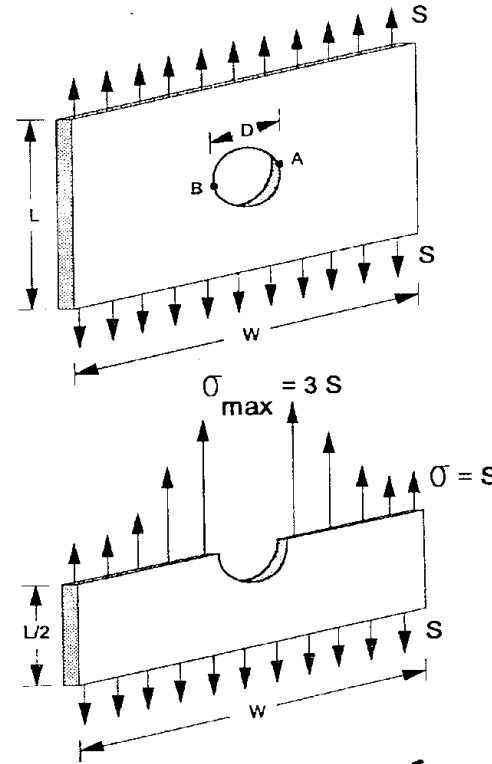
Combination of **Type- 1 & Type- 2**

Example:

On January 28, 1986, the Challenger Space Shuttle exploded because an O-ring seal in one of the main boosters did not respond well to cold weather.

STRESS CONCENTRATION

- ❑ Stress distribution around a hole in an infinite plate was derived by **G. Kirsch** in 1898 using the theory of elasticity
- ❑ The maximum stress is three times the uniform stress
- ❑ $K_t = 3$



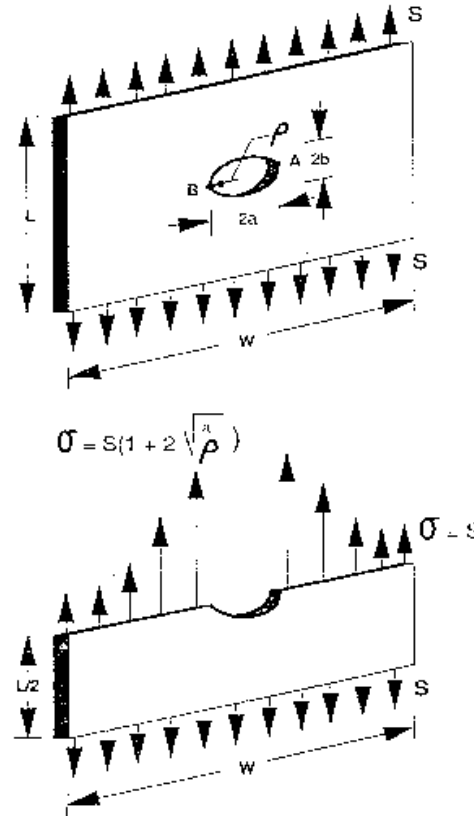
STRESS CONCENTRATION

- ❑ C. E. Inglis (1913) investigated in a plate with an elliptical hole

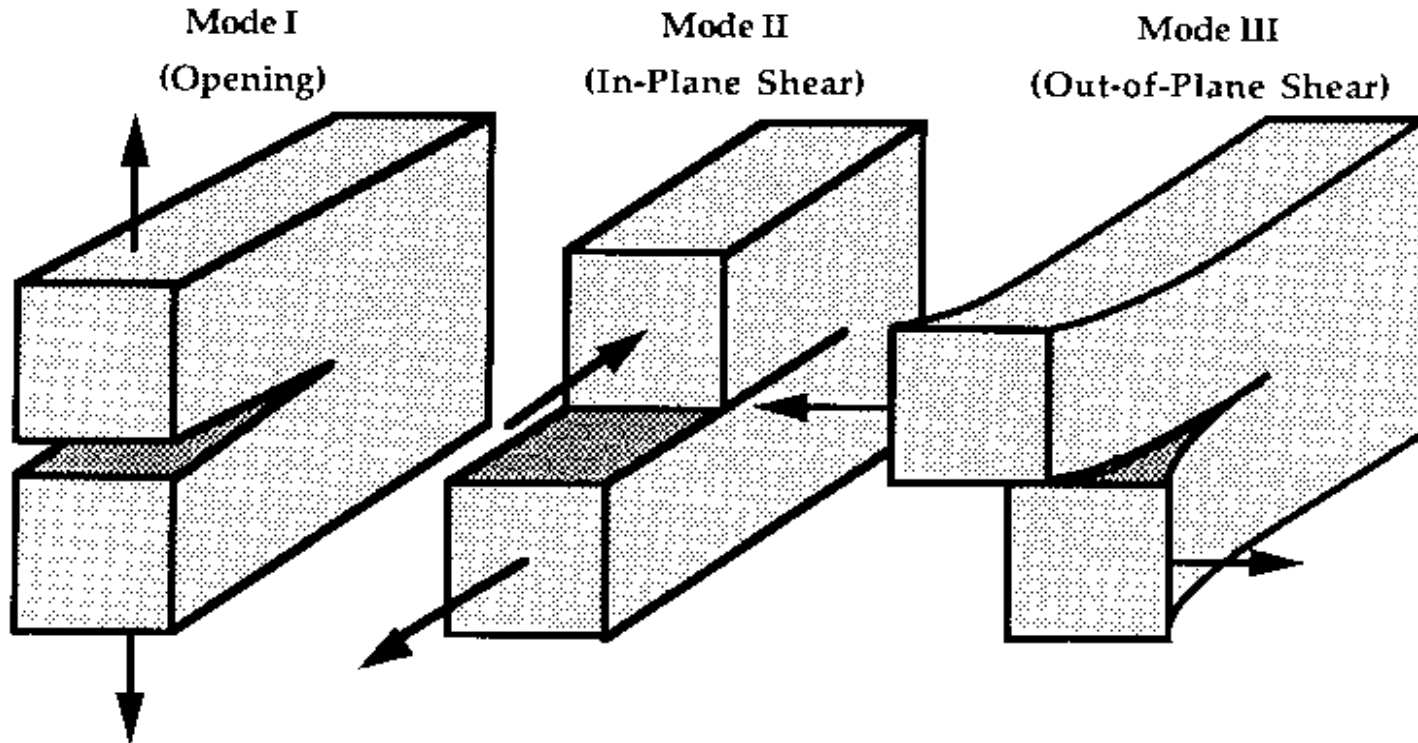
$$K_t = \frac{\sigma_A}{\sigma} = 1 + \frac{2a}{b}$$

$$K_t = 1 + 2\sqrt{\frac{a}{\rho}} \quad \text{where } \rho = \frac{b^2}{a}$$

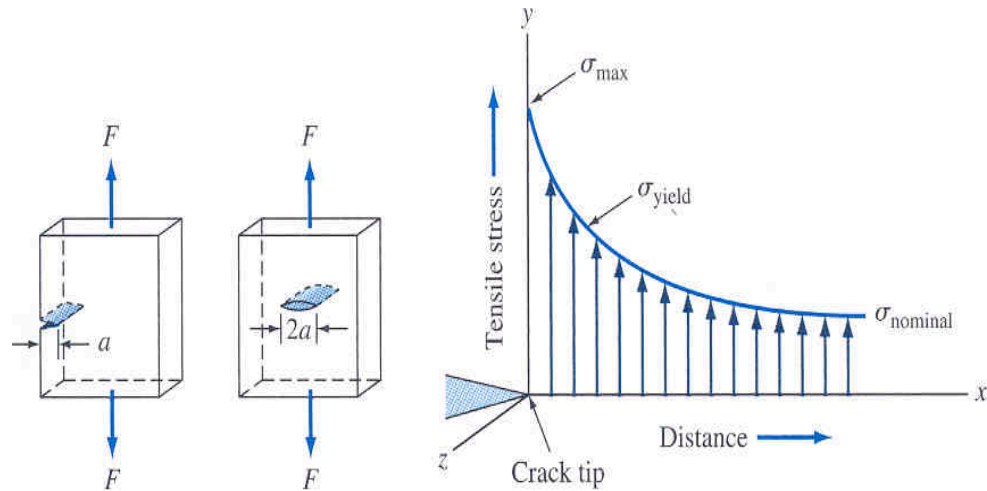
- ❑ Modeling a crack with a ellipse means $\rho \rightarrow 0$;
 $K_t \rightarrow \infty$, infinite stress
- ❑ K_t could not be used for crack problems



Loading Modes



The Stress Intensity Factor



$$K_I = Y \sigma \sqrt{\pi a}$$

K_I = Stress intensity factor.

σ = Applied stress.

a = edge crack length or
half the length of an
internal through crack

Y = geometric constant.

Unit of stress intensity factor K_I is $MPa\sqrt{m}$

Note: Y for center crack infinite plate is 1

Y for edge crack infinite plate is 1.12

Y for finite plates are given by polynomial function

The Stress Intensity Factor

K_{Ic} = critical value of stress intensity factor (**Fracture toughness**)

$$K_{Ic} = Y \sigma_f \sqrt{\pi a}$$

Example:

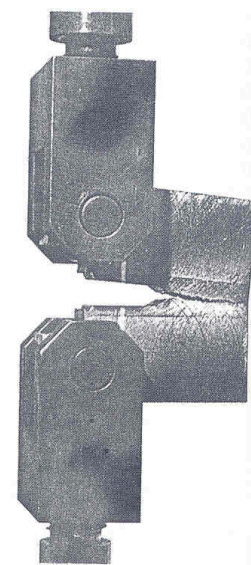
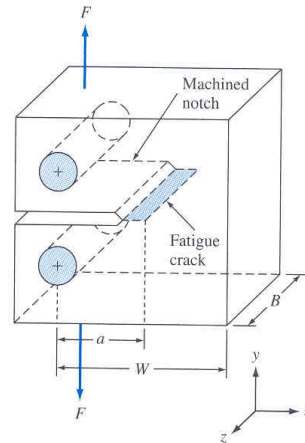
Al 2024 T851	$26.2 \text{ MPa}\sqrt{\text{m}}$
4340 alloy steel	$60.4 \text{ MPa}\sqrt{\text{m}}$

K_{Ic} of Aircraft Materials

Product	Alloy	Temper	Yield strength at 0.2% offset (longitudinal)		Plane-strain fracture toughness, K_{Ic}					
			MPa	ksi	L-T		T-L		S-L	
					MPa√m	ksi√in.	MPa√m	ksi√in.	MPa√m	ksi√in.
Plate	2014	T651	440	64	24	22	22	20	19	17
	2024	T351	325	47	36	33	33	30	26	24
	2024	T851	455	66	24	22	23	21	18	16
	2124	T851	440	64	32	29	25	23	24	22
	2219	T851	435	63	39	35	36	33	...	...
	7050	T73651	455	66	35	32	30	27	29	26
	7075	T651	505	73	29	26	25	23	20	18
	7075	T7651	470	68	30	27	24	22	20	18
	7075	T7351	435	63	32	29	29	26	23	19
	7475	T651	495	72	43	39	37	34	32	29
	7475	T7651	460	67	47	43	39	35	31	28
	7475	T7351	430	62	53	48	42	38	35	32
	7050	T736	455	66	36	33(a)	25	23(b)	25	23(b)
	7149	T73	460	67	34	31(a)	24	22(b)	24	22(b)
Die forgings	7175	T736	490	71	33	30	29	26	29	26
	2024	T852	430	62	29	26	21	19	18	16
	7050	T73652	455	66	36	33	23	21	22	20
Hand forgings	7075	T7352	365	53	37	34	29	26	23	21
	7079	T652	440	64	29	26	25	23	20	18
	7175	T736	470	68	37	34	30	27	26	24
	7050	T7651X	495	72	31	28	26	24	21	19
Extrusions	7050	T7351X	459	65	45	41	32	29	26	24
	7075	T651X	490	71	31	28	26	24	21	19
	7075	T7351X	435	63	35	32	29	26	22	20

(a) Parallel to grain flow. (b) Nonparallel to grain flow

- ❑ A **notch** is machined in a specimen of sufficient thickness B .
- ❑ $B \gg a$ for **plain strain** condition.
- ❑ $B = 2.5 \left(\frac{K_{IC}}{\sigma_{ys}} \right)^2$, where σ_{ys} is yield strength of the material
- ❑ Specimen is tensile tested.
- ❑ Higher the K_{Ic} value, more ductile the metal is.
- ❑ Used in design to find **allowable flaw size**.



ASTM Standard for fracture mechanics

- E616-89 : Terminology Relating to Fracture Testing
- E338-91 : Standard Test Method for Sharp-Notch Tension Testing for High-Strength Sheet Materials
- E399-90 : Standard Test Method for Plane-Strain Fracture Toughness of Metallic Materials
- E561-94 : Practice for R-Curve Determination
- E813-89 : Standard Test Method for J_{IC} , A Measure of Fracture Toughness
- E1152-87 : Standard Test Method for Determining J-R Curve
- E1304-89 : Standard Test Method for Plane-Strain (Chevron-Notch) Fracture Toughness of Metallic Materials
- E1290-93 : Standard Test Method for Crack-Tip Opening Displacement (CTOD) Fracture Toughness Measurement
- E1150-87 : Definition of Terms Relating to Fatigue
- E647-93 : Standard Test Method for Measurement of Fatigue Crack Growth Rate
- D5045-91a : Standard Test Method for Plane-Strain Fracture Toughness and Strain Energy Release Rate of Plastic Materials

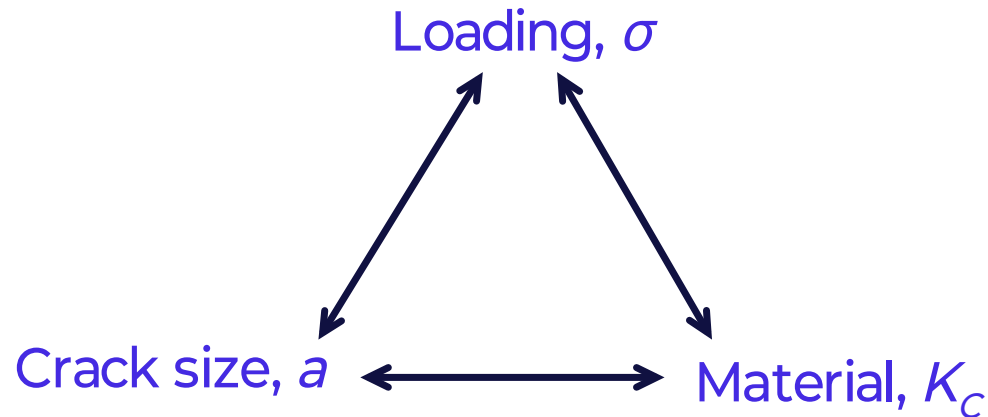


Importance of Fracture Mechanics

1. Selection of a Material
2. Design of a Component
3. Design of a manufacturing or Testing Method

Design based on Fracture Mechanics

$$K_c = Y \sigma_c \sqrt{\pi a_c}$$



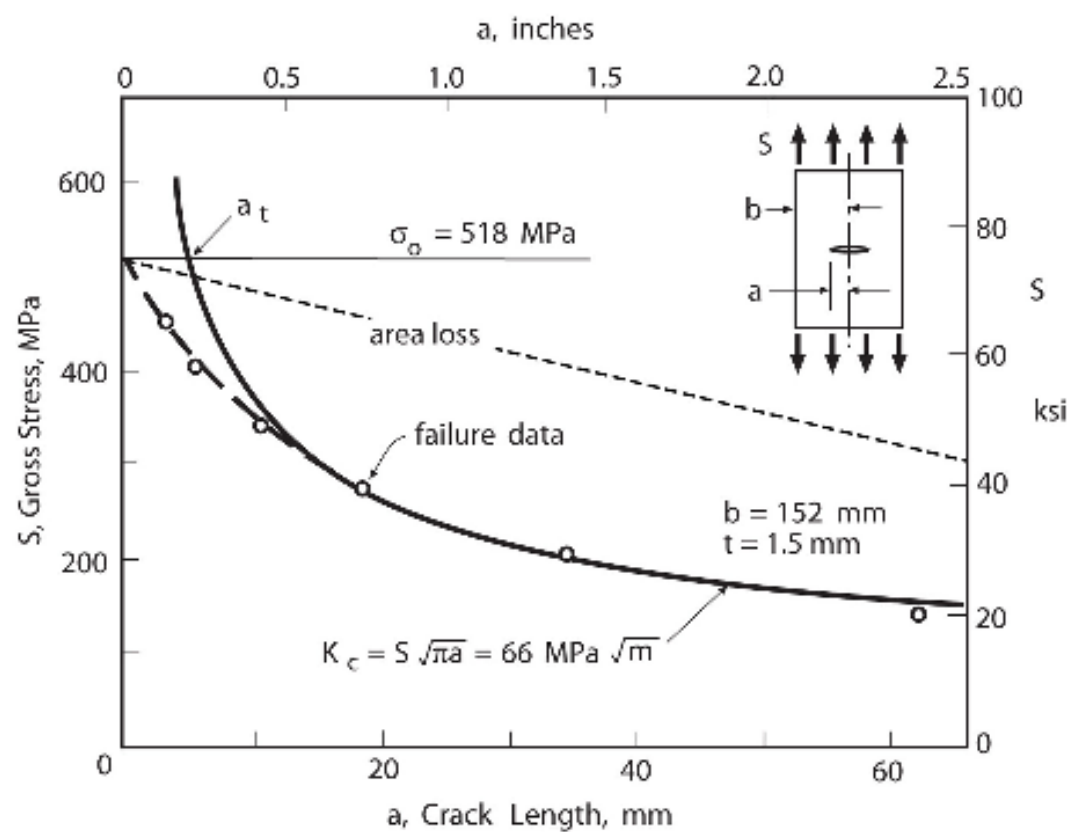
Comparison between conventional and the Damage tolerant design

Conventional Design

- ☐ Based on Factor of safety
- ☐ Conservative Design
- ☐ Normally do not permit flaw in the components
- ☐ Costlier design
- ☐ Not much risk involved

Damage Tolerant design

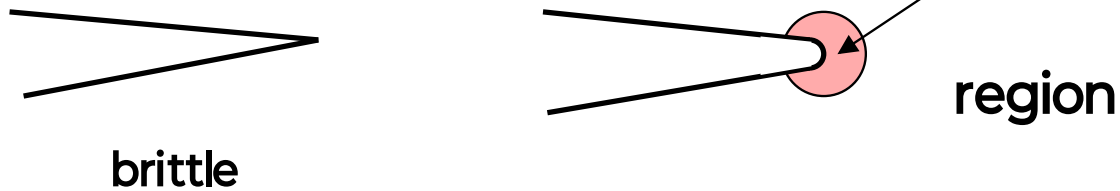
- ☐ Based on Fracture toughness
- ☐ Optimised design
- ☐ Permits flaw within prescribed limit
- ☐ Cheap design
- ☐ Calculated risk involved



Crack Propagation

Cracks propagate due to sharpness of crack tip

- A plastic material deforms at the tip, “blunting” the crack.



Energy balance on the crack

- Elastic strain energy-
 - energy is stored in material as it is elastically deformed
 - this energy is released when the crack propagates
 - creation of new surfaces requires (this) energy

Design based on Fracture Mechanics

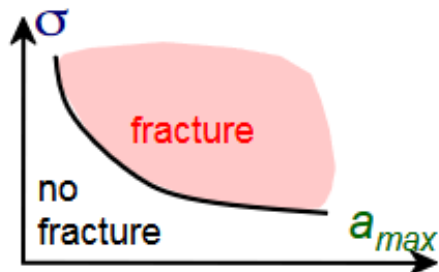
- Crack growth condition:

$$K \geq K_c = Y\sigma\sqrt{\pi a}$$

- Largest, most stressed cracks grow first!

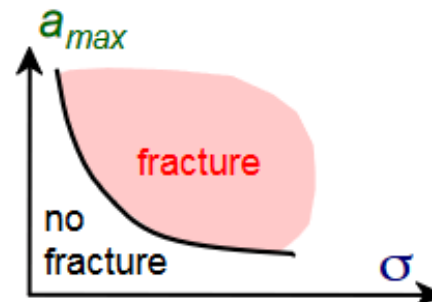
--Result 1: Max. flaw size dictates design stress.

$$\sigma_{design} < \frac{K_c}{Y\sqrt{\pi a_{max}}}$$



--Result 2: Design stress dictates max. flaw size.

$$a_{max} < \frac{1}{\pi} \left(\frac{K_c}{Y\sigma_{design}} \right)^2$$



Design based on Fracture Mechanics

Design Example: Aircraft Wing

Material has $K_c = 26 \text{ MPa}\sqrt{m}$

Two designs to consider...

Design A

--largest flaw is 9 mm

--failure occurs at stress = 112 MPa

Use....
$$\sigma_c = \frac{K_c}{Y \sqrt{\pi a_c}}$$

Design B

--use same material

--largest flaw is 4 mm

--failure stress= ?

Key point: Y and K_c are the same in both designs!

$$(\sigma_c \sqrt{\pi a_{\max}})_A = (\sigma_c \sqrt{\pi a_{\max}})_B$$

Answer $(\sigma_c)_B = 168 \text{ MPa}$

Reducing flaw size pays

off!



Fracture Toughness - K_C , K_{IC}

- ❑ Higher yield or ultimate strength generally produces a decrease in K_{IC} , and thus a greater susceptibility for catastrophic fracture!
- ❑ Low impurity materials provide better fracture toughness.
- ❑ K_{IC} can be very sensitive to metallurgical conditions such as grain orientation, chemistry, and microstructure.
- ❑ Fracture toughness K_C of metals is also dependent on temperature, strain rate, and corrosive environment.

Prob.1

A structural plate component of an engineering design must support 207 MPa in tension. If aluminum alloy 2024-T851 is used for this application, what is the largest internal flaw size that this material can support? (Use $Y = 1$).

■ Solution

$$K_{IC} = Y\sigma_f \sqrt{\pi a} \quad (7.2)$$

Using $Y = 1$ and $K_{IC} = 26.4 \text{ MPa}\sqrt{\text{m}}$ from Table 7.1,

$$a = \frac{1}{\pi} \left(\frac{K_{IC}}{\sigma_f} \right)^2 = \frac{1}{\pi} \left(\frac{26.4 \text{ MPa}\sqrt{\text{m}}}{207 \text{ MPa}} \right)^2 = 0.00518 \text{ m} = 5.18 \text{ mm}$$

Thus, the largest internal crack size that this plate can support is $2a$, or $(2)(5.18 \text{ mm}) = 10.36 \text{ mm}$.



References

1. William F. Smith, Javad Hashemi, 'Material Science & Engineering", In SI units Fourth edition, McGrawHill.
2. T. L. Anderson, Fracture Mechanics; Fundamentals and Applications.

Thank You



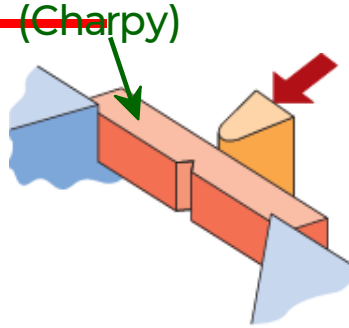
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Pilani Campus

Impact and Hardness testing (DE G 631) Dated: 10/09/2026

Prof .P. Srinivasan

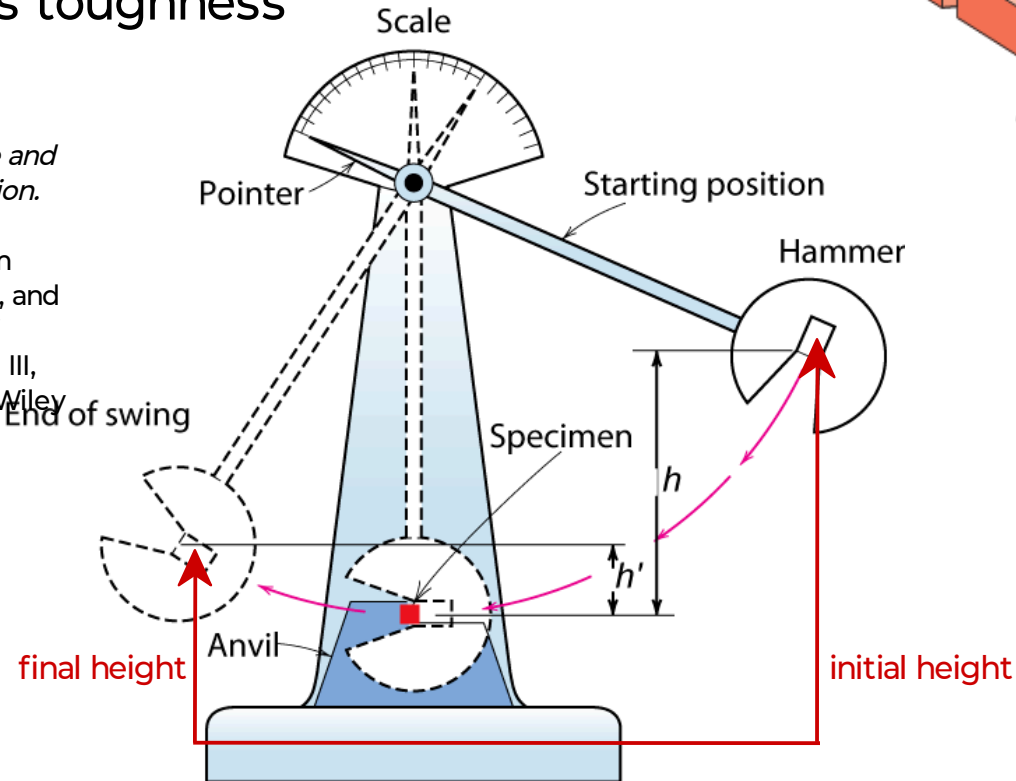
Impact Testing

- Impact loading:
 - severe testing case
 - makes material more brittle
 - decreases toughness



Adapted from Fig. 11.12(b),
Callister's Materials Science and Engineering, Adapted Version.

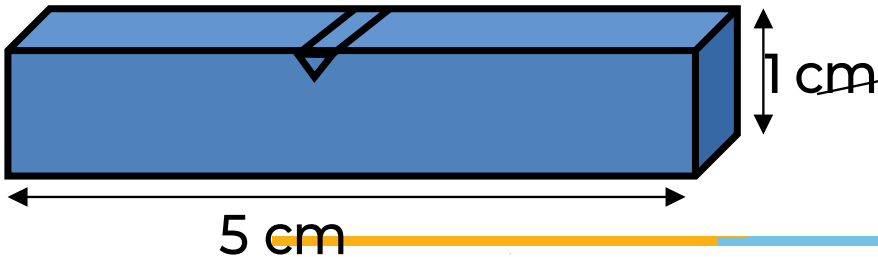
(Fig. 11.12(b) is adapted from
 H.W. Hayden, W.G. Moffatt, and
 J. Wulff, *The Structure and
 Properties of Materials*, Vol. III,
Mechanical Behavior, John Wiley
 and Sons, Inc. (1965) p. 13.)





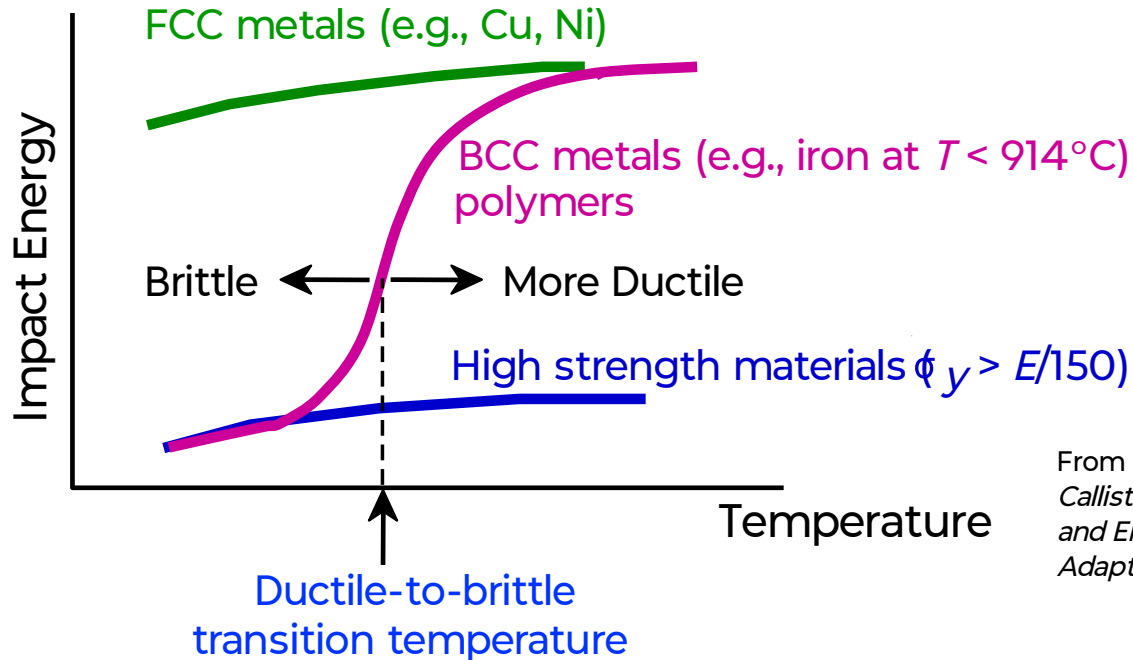
The Set-up

- Procedure - ASTM E23
- Material - Cold Rolled 1018
- Test Temperatures
 1. Boiling Water (100 °C)
 2. Room Temp (~ 25 °C)
 3. Ice Water (0)
 4. Dry Ice (-78 °C)
 5. Liquid Nitrogen (-196 °C)

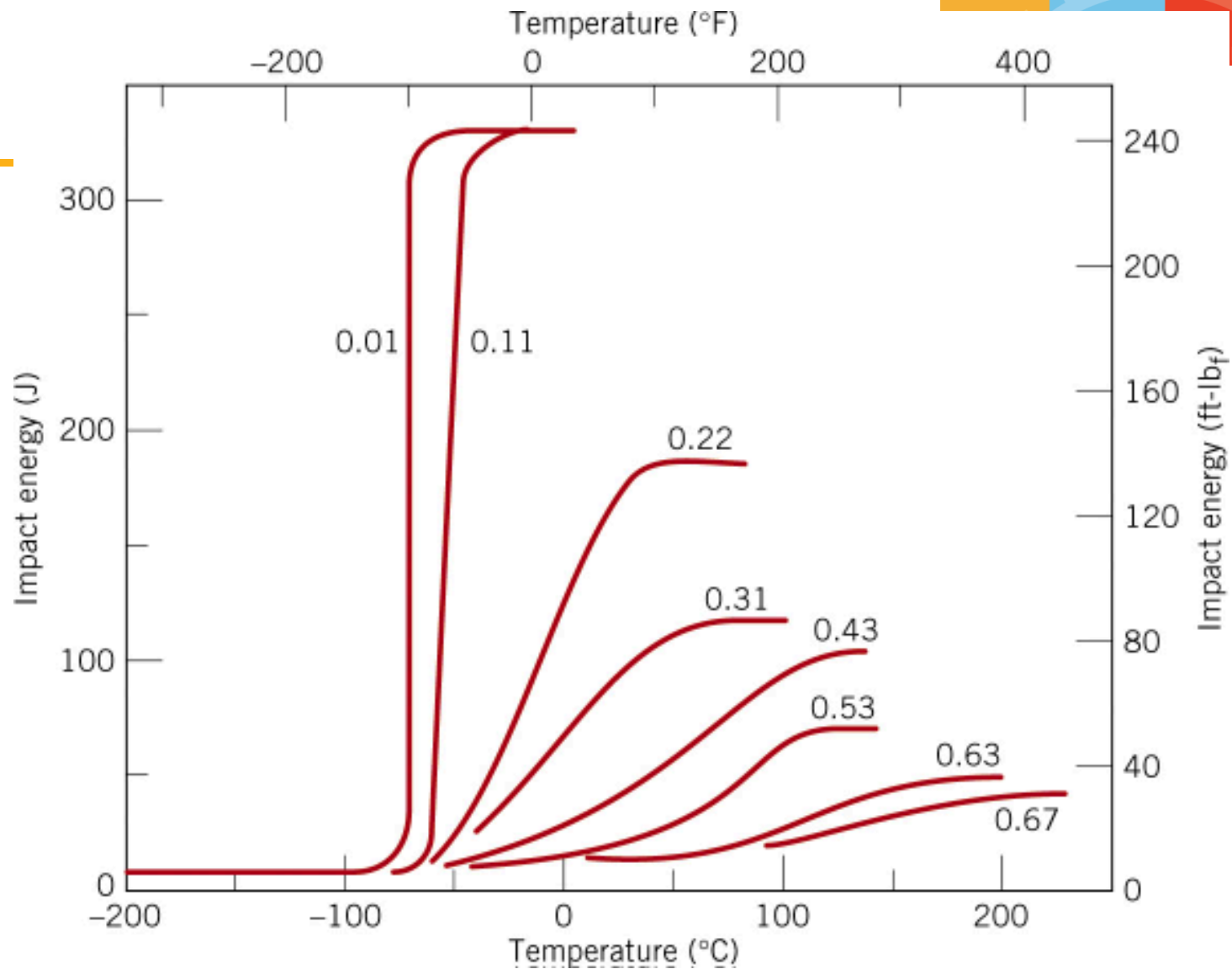


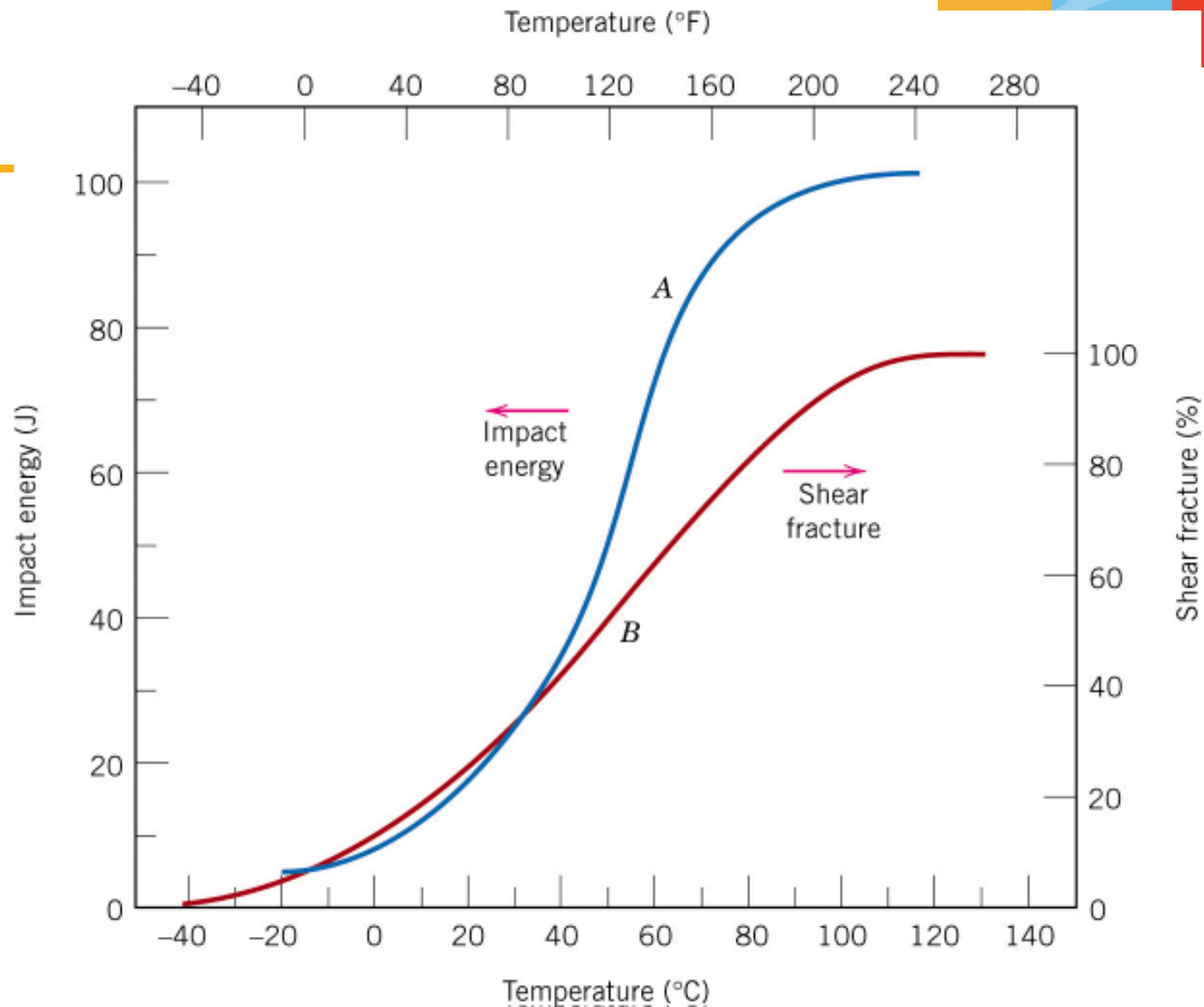
Temperature

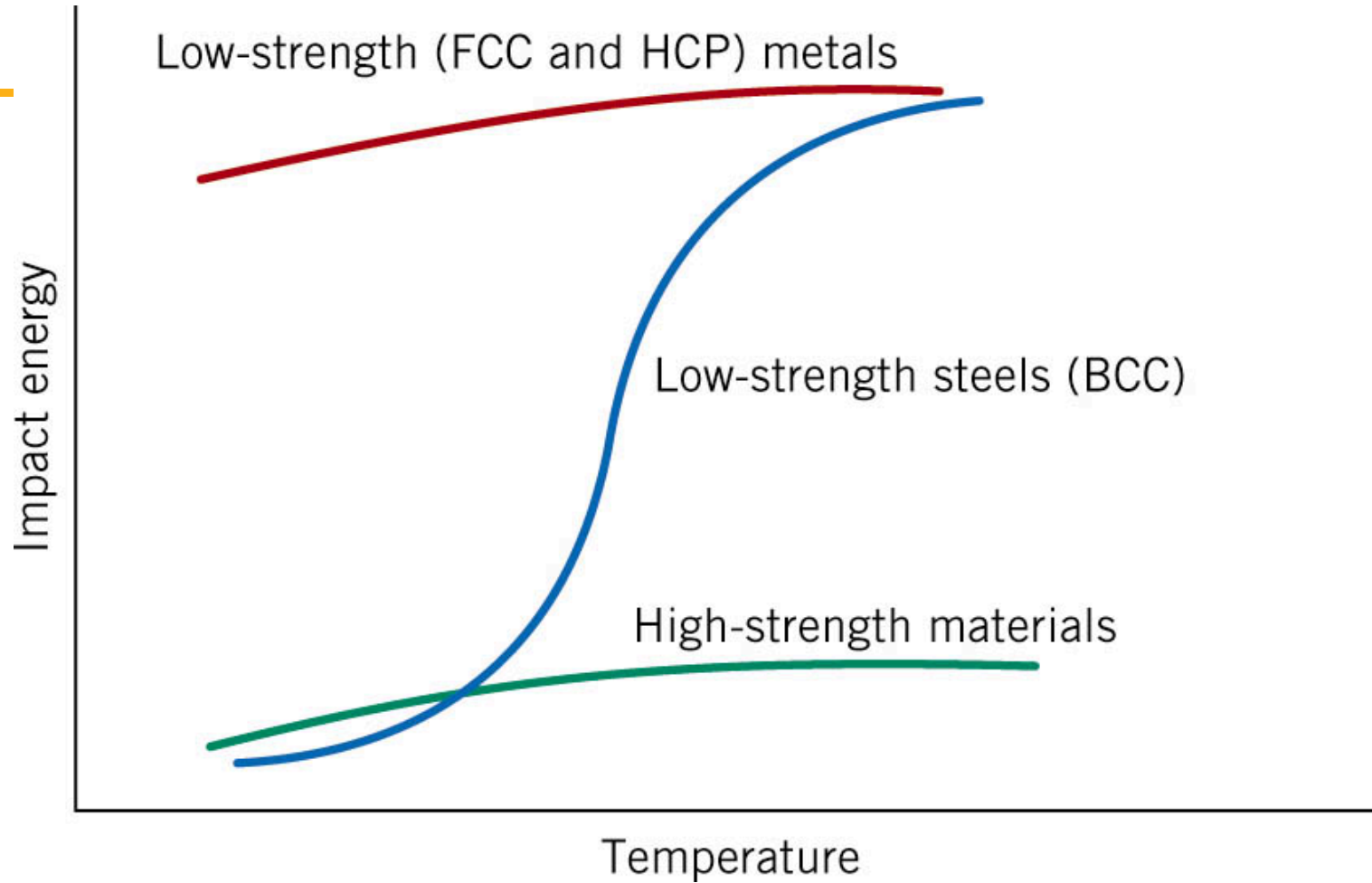
- Increasing temperature...
 - increases % EL and K_c
- Ductile-to-Brittle Transition Temperature (DBTT)...

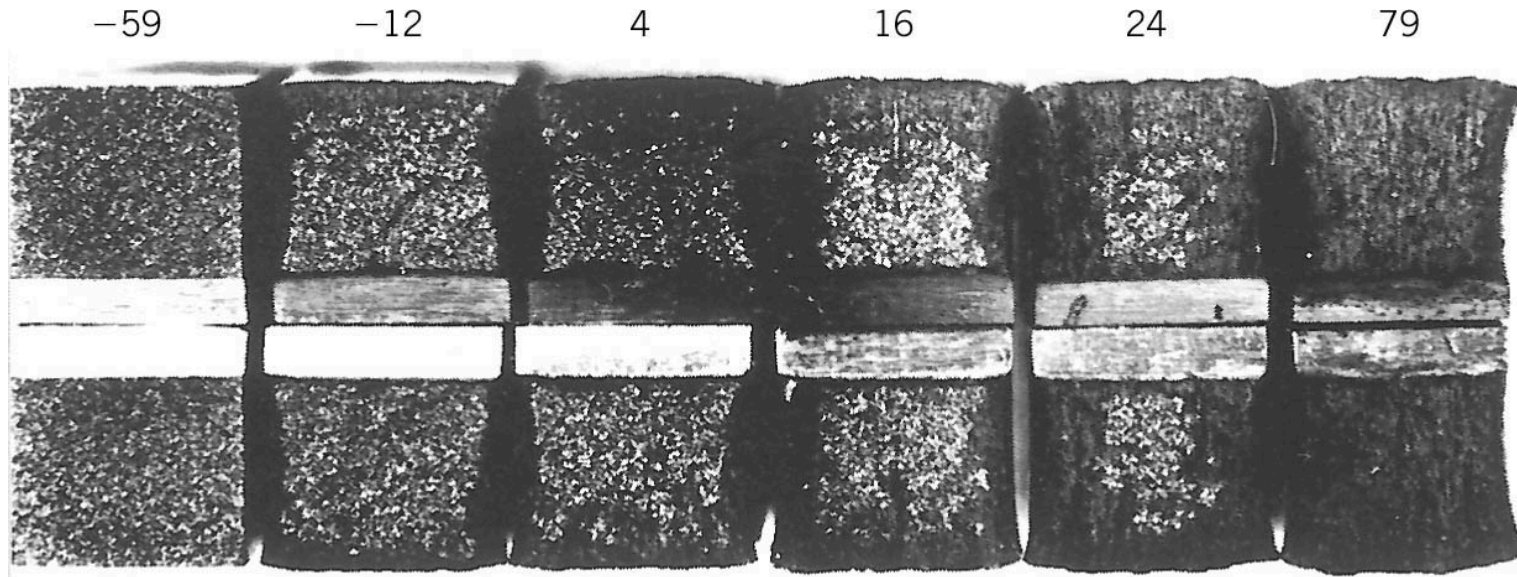


From Fig. 11.15
*Callister's Materials Science
 and Engineering,
 Adapted Version.*









f14_1l_pg405

Design Strategy: Stay Above The DBTT!

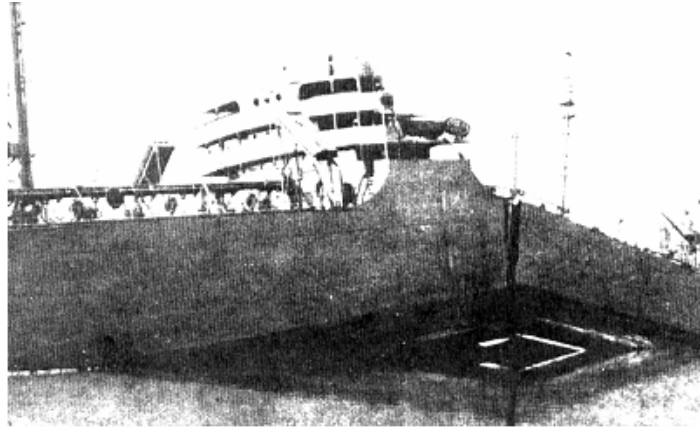


- Pre-WWII: The Titanic



Reprinted w/ permission from R.W. Hertzberg , "Deformation and Fracture Mechanics of Engineering Materials", (4th ed.) Fig. 7.1(a), p. 262, John Wiley and Sons, Inc., 1996. (Orig. source: Dr. Robert D. Ballard, *The Discovery of the Titanic*.)

- WWII: Liberty ships



Reprinted w/ permission from R.W. Hertzberg , "Deformation and Fracture Mechanics of Engineering Materials", (4th ed.) Fig. 7.1(b), p. 262, John Wiley and Sons, Inc., 1996. (Orig. source: Earl R. Parker, "Behavior of Engineering Structures", Nat. Acad. Sci., Nat. Res. Council, John Wiley and Sons, Inc ., NY, 1957.)

- Problem: Used a type of steel with a DBTT ~ Room temp.



Following is tabulated data that were gathered from a series of Charpy impact tests on a commercial low-carbon steel alloy.

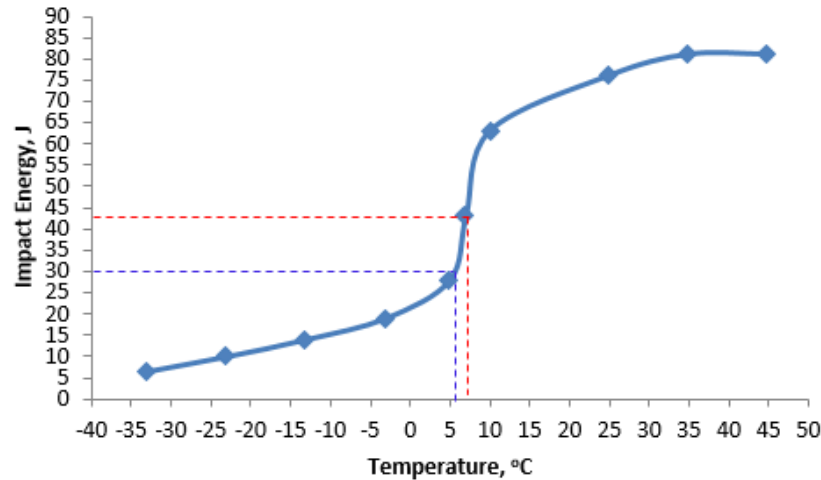
Temperature (°C)	Impact Energy (J)
45	81
35	81
25	76
10	63
7	43
5	28
-3	19
-13	14
-23	10
-33	6.5

(a) Plot the data as impact energy versus temperature.

(b) Determine a ductile-to-brittle transition temperature as that temperature corresponding to the average of the maximum and minimum impact energies.

(c) Determine a ductile-to-brittle transition temperature as that temperature at which the impact energy is 30 J.

a) Plot of impact energy versus temperature



b) Average of the maximum and minimum impact energies is $(81 + 6.5)/2 = 43.75$ J. Temperature corresponding to 43.75J is 7°C.

c) Temperature corresponding to 30J is 5.5°C.



Cryogenic Temperatures

What is Cryogenics?

Cryogenics is the science and technology associated with processes occurring below about 120 K. In particular, this includes refrigeration, liquefaction, storage and transport of cryogenic fluids.

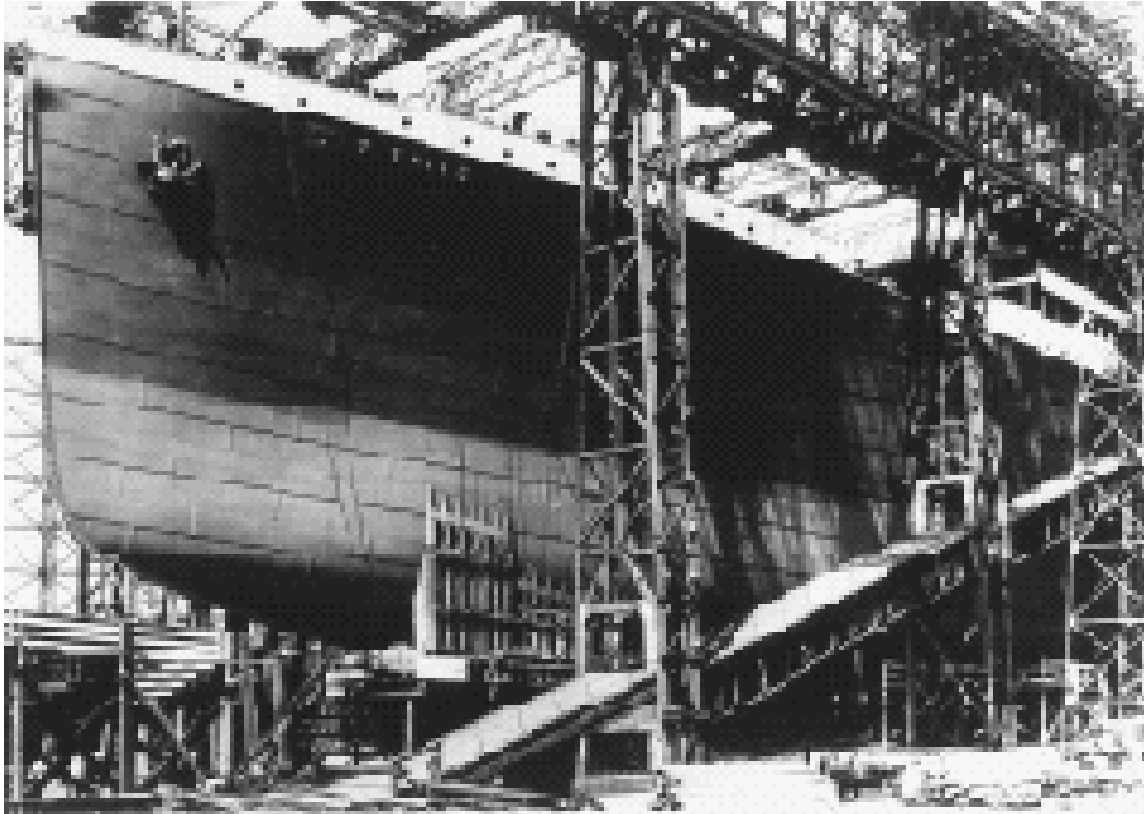
The Kelvin Temperature Scale

$$K = ^\circ C + 273$$

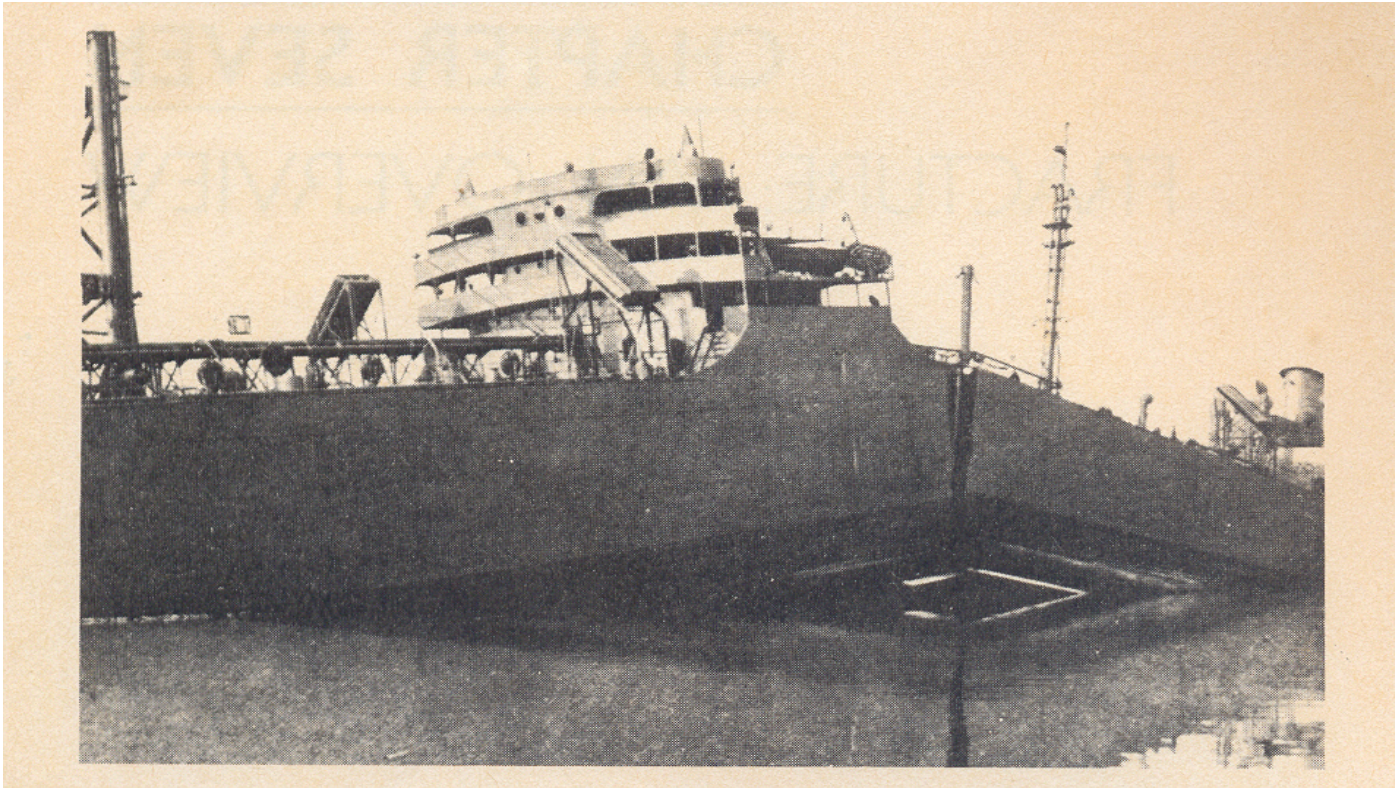
Room temperature ~ 300 K

LN ₂	77 K (-196 °C)
LH ₂	20 K (-253 °C)
LHe	4.2 K (-269°C)

Titanic ship under construction



Fracture of ship due to Brittle fracture



Composition of Steels taken from failed Titanic ship



Gate, and ASTM A36 Steel

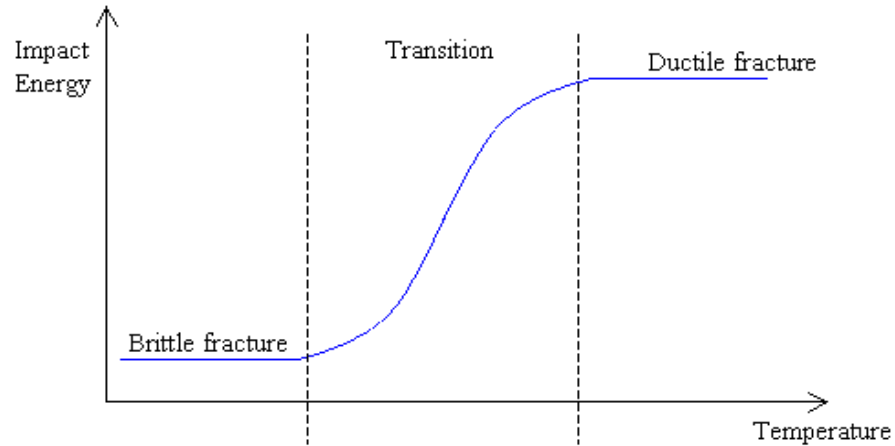
	C	Mn	P	S	Si	Cu	O	N	MnS: Ratio
<i>Titanic</i> Hull Plate	0.21	0.47	0.045	0.069	0.017	0.024	0.013	0.0035	6.8:1
Lock Gate*	0.25	0.52	0.01	0.03	0.02	—	0.018	0.0035	17.3:1
ASTM A36	0.20	0.55	0.012	0.037	0.007	0.01	0.079	0.0032	14.9:1

*Steel from a lock gate at the Chittenden ship lock between Lake Washington and Puget Sound, Seattle, Washington.

Charpy test



Ductile to Brittle transformation Temperature phenomenon





Factors influencing ductile to brittle transition temperature

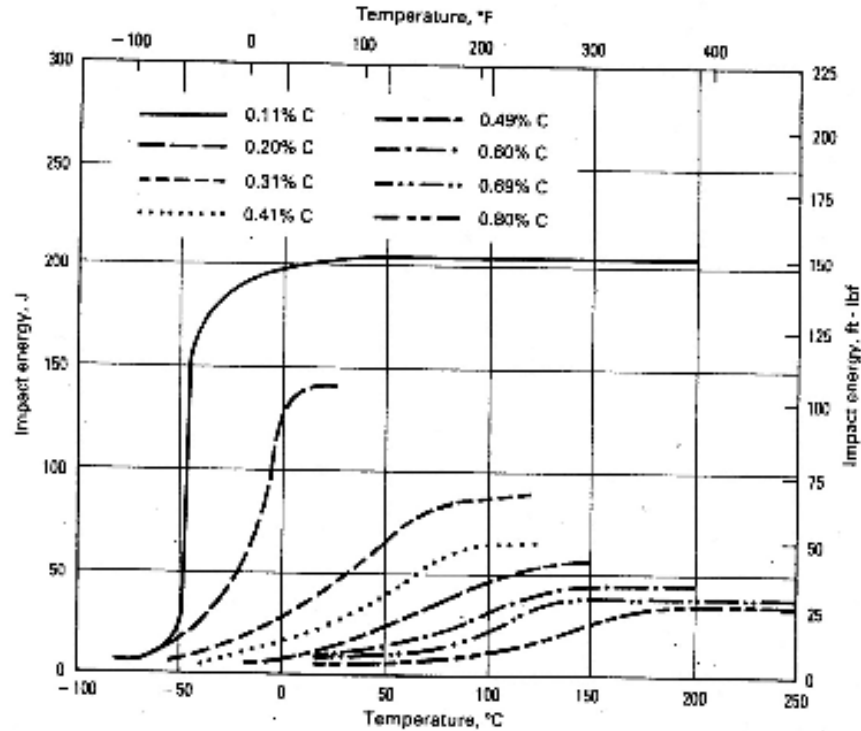
- Composition of steel.
- Grain size
- Heat treatment
- Micro structure
- Thermo mechanical treatment

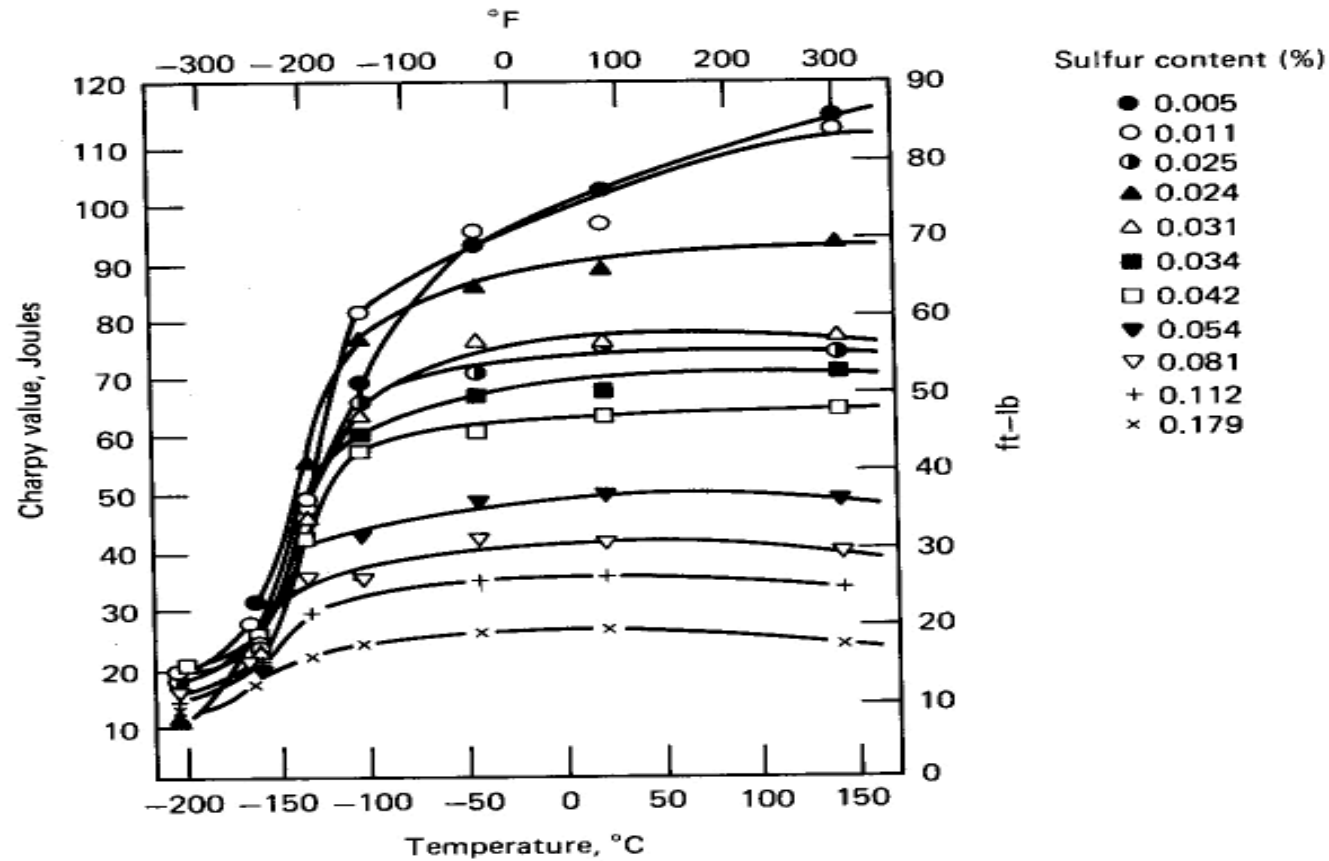


Important elements and it's effect

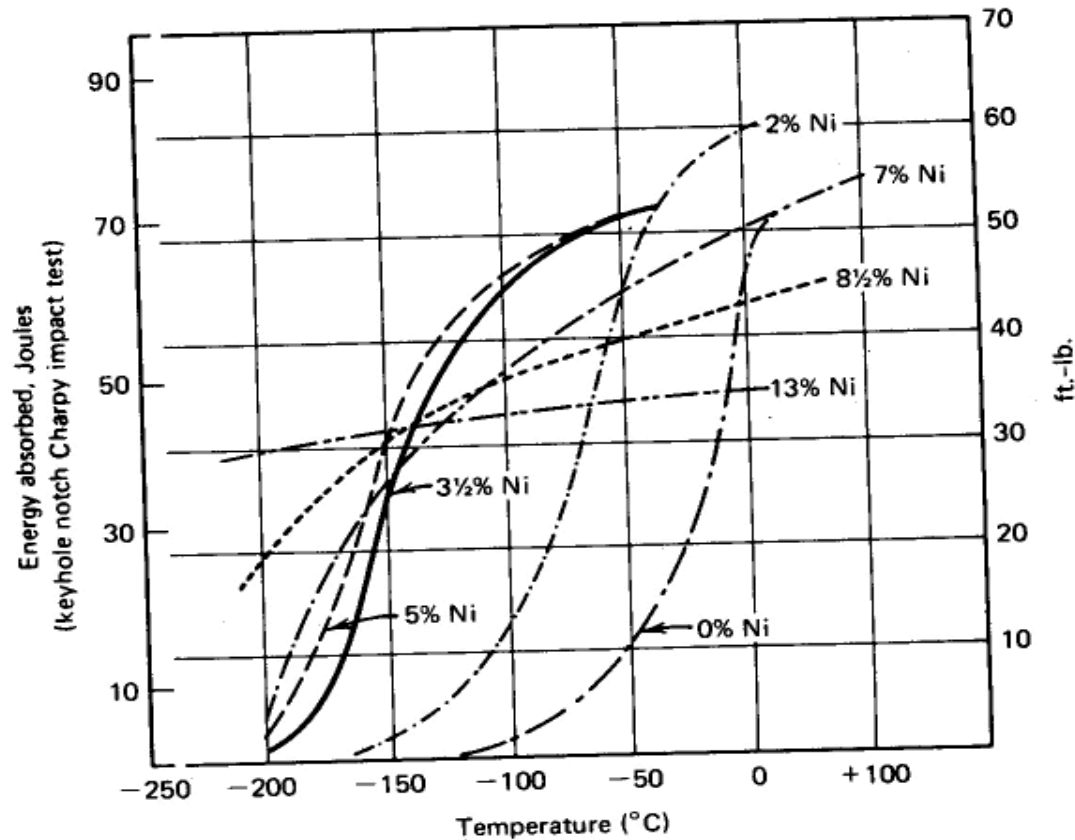
- Carbon
- Manganese
- Sulfur
- Phosphorus
- Nickel
- Niobium

Effect of Carbon and DBTT

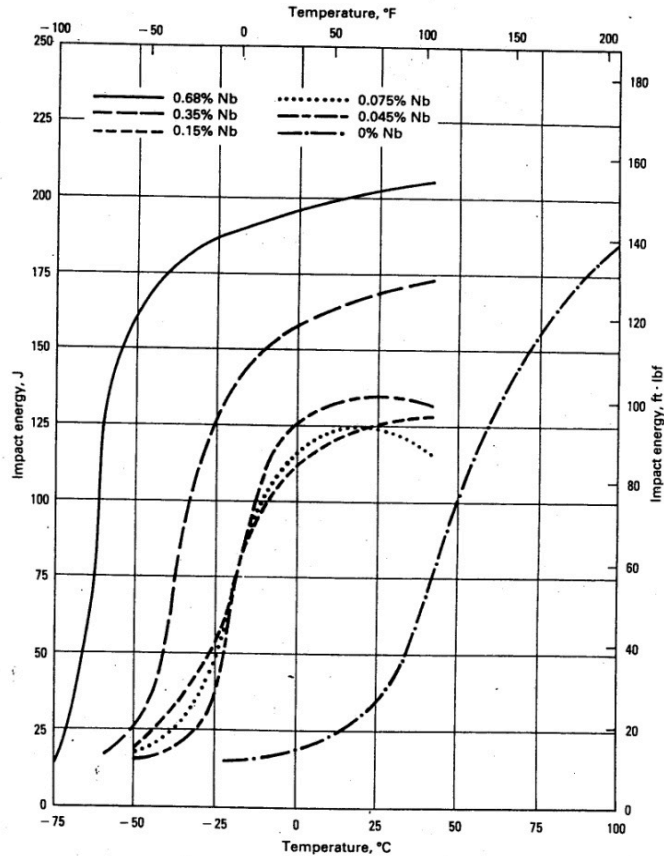




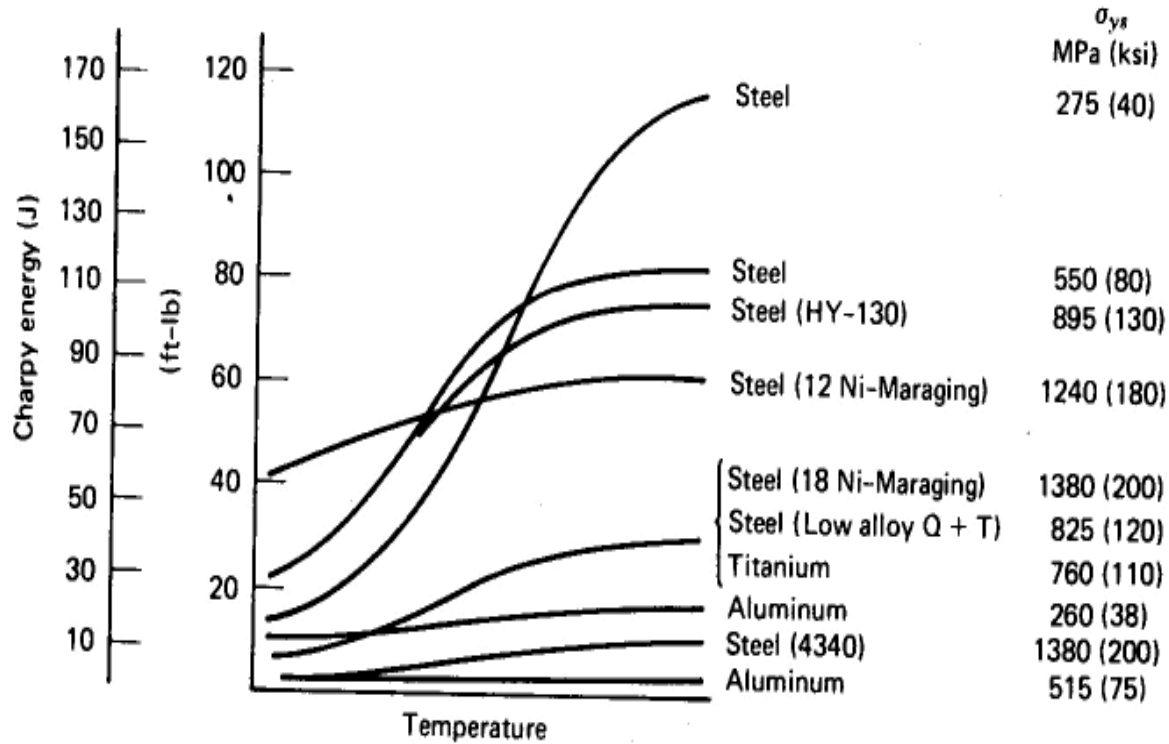
Effect of Nickel and DBTT



Effect of Niobium and DBTT



Effect of Composition of steel and DBTT



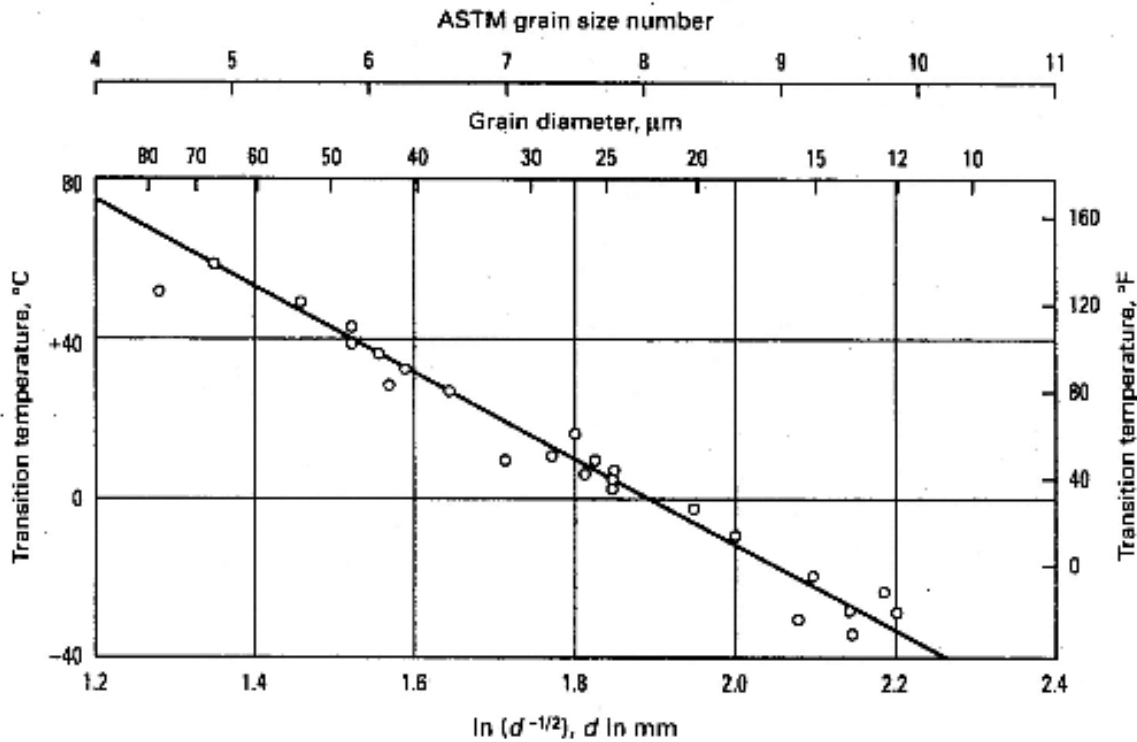


Effect of Grain Size and heat

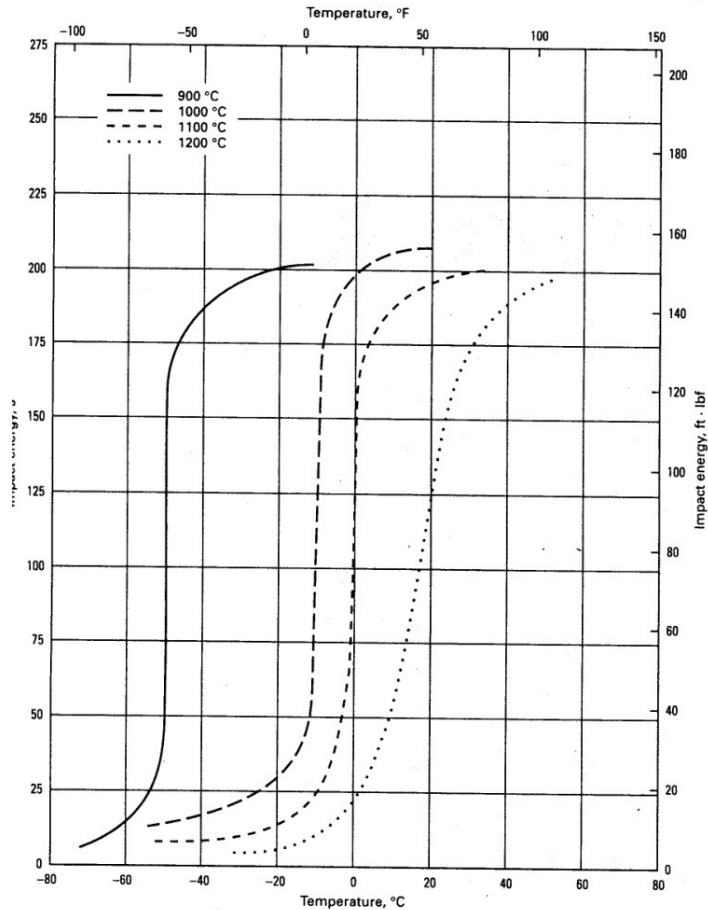
treatment
Fine grain structure will have low DBTT

- Hot deformation process and secondary heat treatment is done to provide fine grains.

Effect of grain size in transition temperature



Effect of Normalizing Temperature

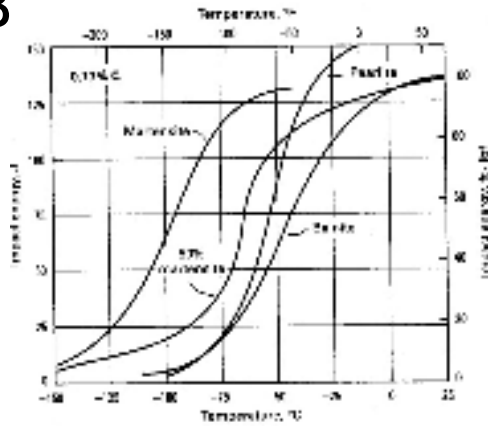




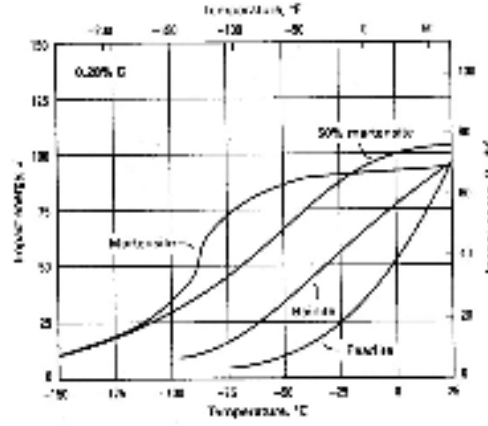
Effect of phases present

- Pearlite and Ferrite have high DBTT
- Upper Bainite possesses relatively low DBTT
- Tempered martensite is considered to be better than pearlite and bainite.

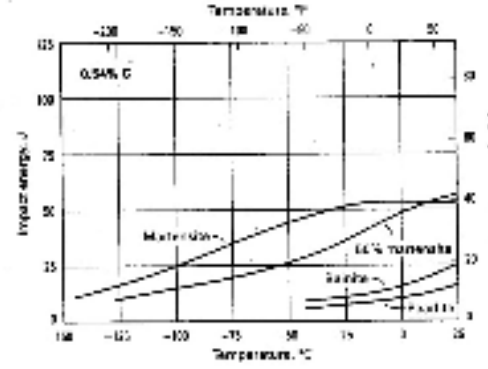
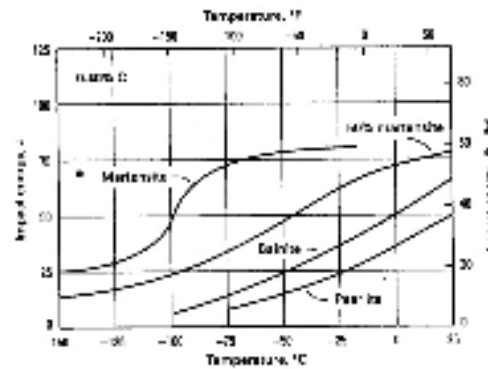
Effect of Carbon and Phase of steel in DB



(a)



(b)





Thermo mechanical treatment and Maraging steel

- Retained Austenite and subsequent transformation induced plasticity drastically increases the fracture toughness.
- This processing is done under high precision resulting in manufacture of Maraging steels.

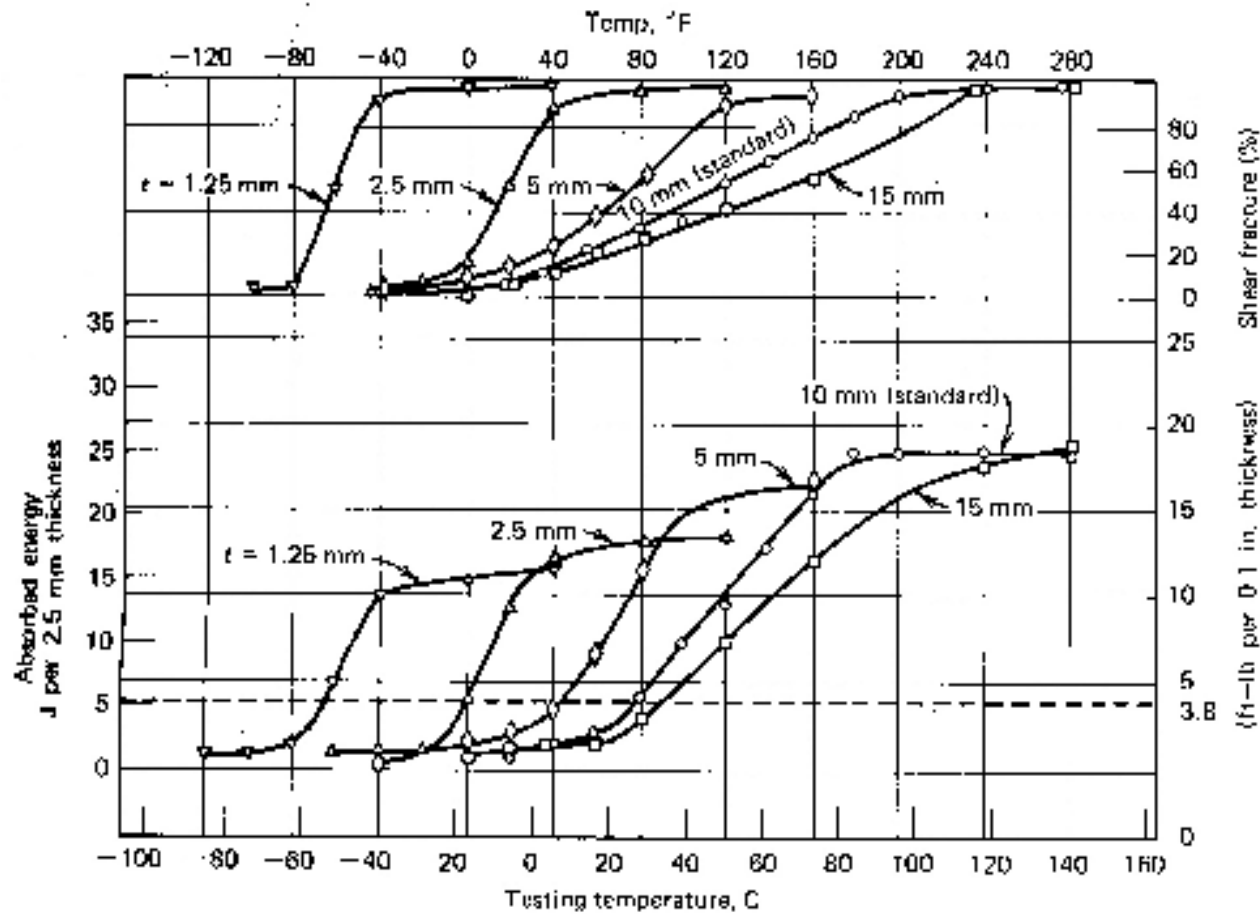
Maraging steels

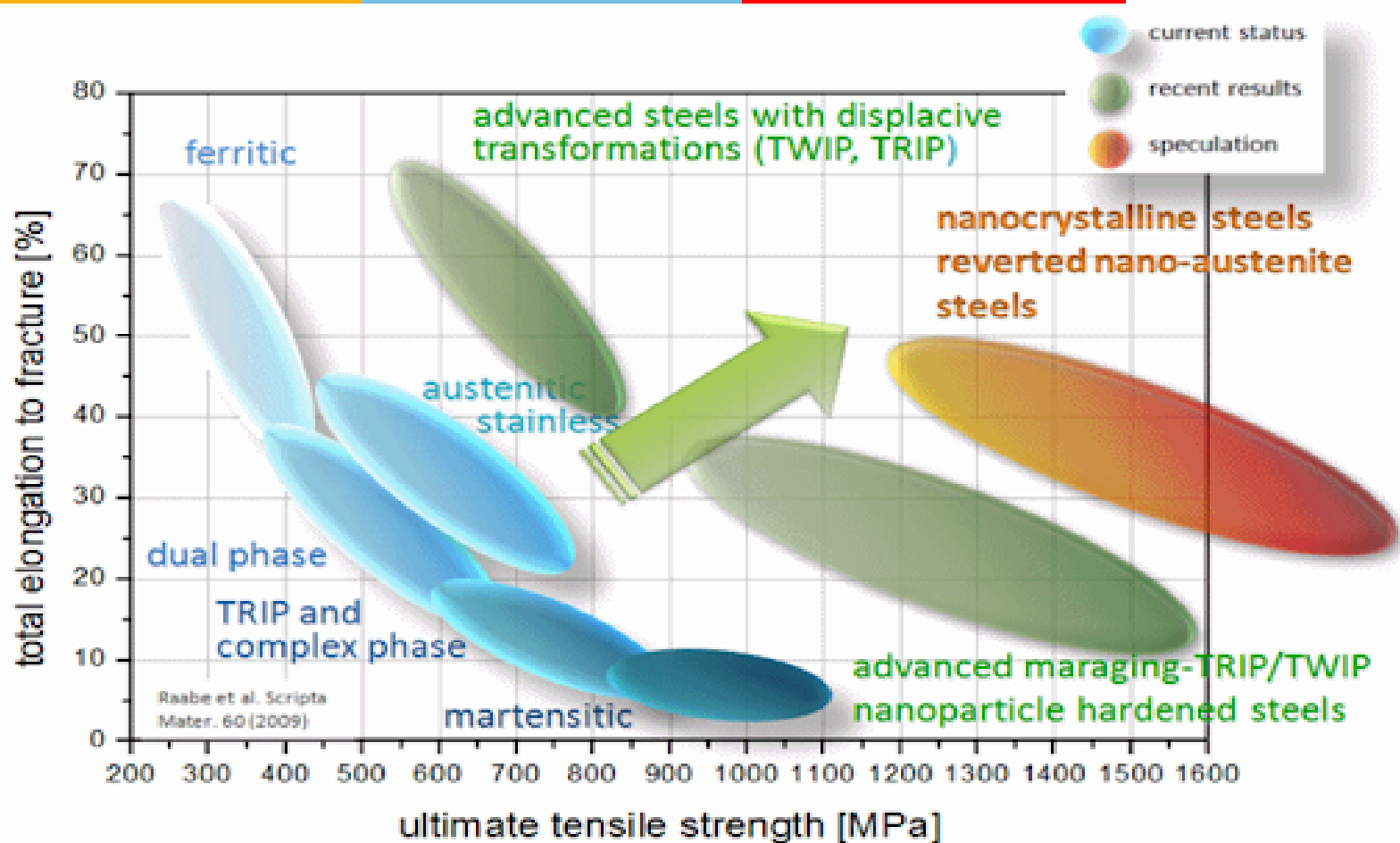
- On cooling from the austenitic condition the alloy transforms to a fine lath type martensite, and precipitation hardening is induced by "maraging" at 480 °C. Many types of precipitates have been reported (e.g. Ni_3TiAl) but the main hardener is probably orthorhombic Ni₃Mo, the solubility of which is probably reduced by Cobalt. The steels have high fracture toughness, K_{Ic} due to a combination of fine grain size of the martensite and the high dislocation density, leading to fine precipitation.



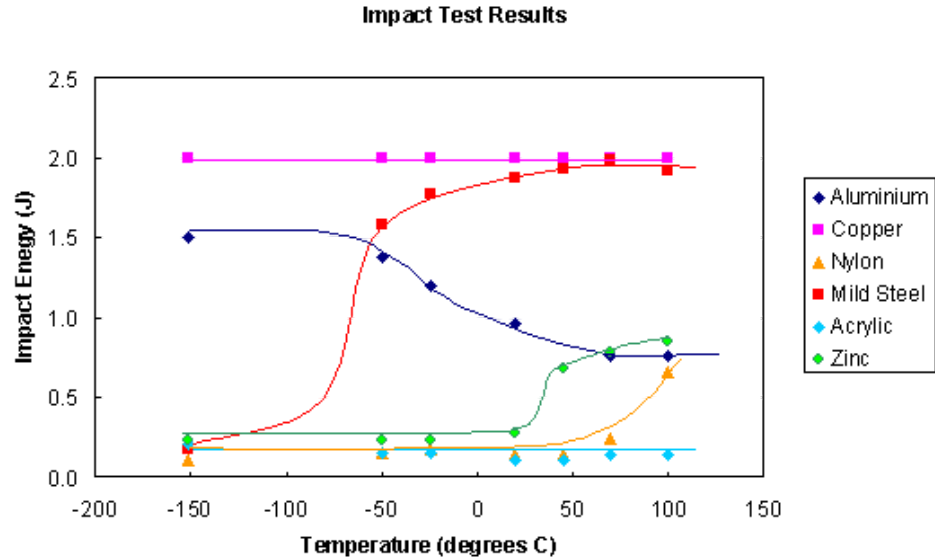
Limitation of DBTT approach

- Results obtained varies significantly with the thickness
- Notch length has pronounced influence





Ductile to Brittle Transition Temperature (DBTT) of various materials





Aluminum alloys for cryogenic applications

- Not much change in properties over wide range of temperatures.
- Used in structural parts at 4 K
- Alloy 6061-T651 for temperature of 77K
- Do not have the limitation of DBTT
- High strength to weight ratio

Examples of Suitable Materials for Cryogenics

- Maraging steels
- Austenitic stainless steels e.g. 304, 304L, 316, 321
- Aluminum alloys e.g. 6061, 6063, 1100
- Copper e.g. OFHC, ETP and phosphorous deoxidized
- Brass
- Fiber reinforced plastics such as G -10 and G -11
- Niobium & Titanium (frequently used in superconducting RF systems)
- Invar (Ni /Fe alloy) useful in making washers due to its lower coefficient of expansion
- Indium (used as an O ring material)
- Kapton and Mylar (used in Multilayer Insulation and as electrical insulation)
- Quartz (used in windows)

Some Unsuitable Materials

- Martensitic stainless steels - Undergoes ductile to brittle transition when cooled down.
- Cast Iron – also becomes brittle
- Carbon steels – also becomes brittle. Sometimes used in 300 K vacuum vessels but care must be taken that breaks in cryogenic lines do not cause the vacuum vessels to cool down and fail
- Rubber, Teflon and most plastics although plastic insulated wires are frequently OK as long as the wire is not repeatedly flexed which could lead to cracking of the insulation



Conclusion

- Ductile to Brittle transition temperature is very important tool for selection of materials for cryogenic application
- This has it's own limitation due to various uncertainties
- Proper selection of composition, heat treatment, phases, mechanical treatment will give better results.



References

1. George E. Dieter, 'Mechanical Metallurgy', Mc GrawHill



BITS Pilani
Pilani Campus

Material Technology & Testing (DE ZG631)

Prof . P. Srinivasan



Introduction to Fracture Mechanics



Fracture Mechanics

Introduction

Why Structures Fail

- ❑ Negligence during design, construction, or operation of the structure ----- **Type- 1**
- ❑ Application of a new design or material, which produces an unexpected (and undesirable) result ----- **Type- 2**



Fracture Mechanics

Introduction

- ❑ Negligence during design, construction, or operation of the structure ----- **Type- 1**

due to human error, ignorance, or willful misconduct. Poor workmanship, inappropriate or substandard materials, errors in stress analysis, and operator error

Example: Boston Molasses Tank Failure



Fracture Mechanics

Introduction

- ❑ Application of a new design or material, which produces an unexpected (and undesirable) result ----- **Type- 2**

Improved design is introduced however there are factors that the designer does not anticipate.

New materials can offer tremendous advantages, but also potential problems.

Example: Liberty Ship Failure, Comet Disaster



Introduction

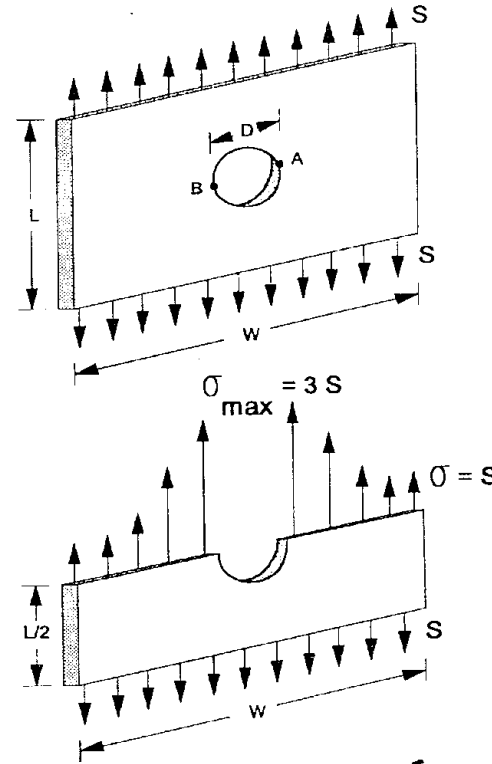
Combination of **Type- 1 & Type- 2**

Example:

On January 28, 1986, the Challenger Space Shuttle exploded because an O-ring seal in one of the main boosters did not respond well to cold weather.

STRESS CONCENTRATION

- ❑ Stress distribution around a hole in an infinite plate was derived by **G. Kirsch** in 1898 using the theory of elasticity
- ❑ The maximum stress is three times the uniform stress
- ❑ $K_t = 3$



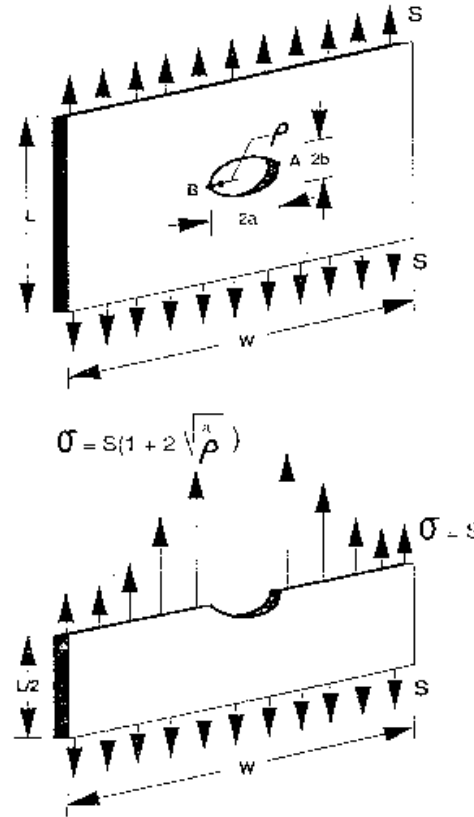
STRESS CONCENTRATION

- ❑ C. E. Inglis (1913) investigated in a plate with an elliptical hole

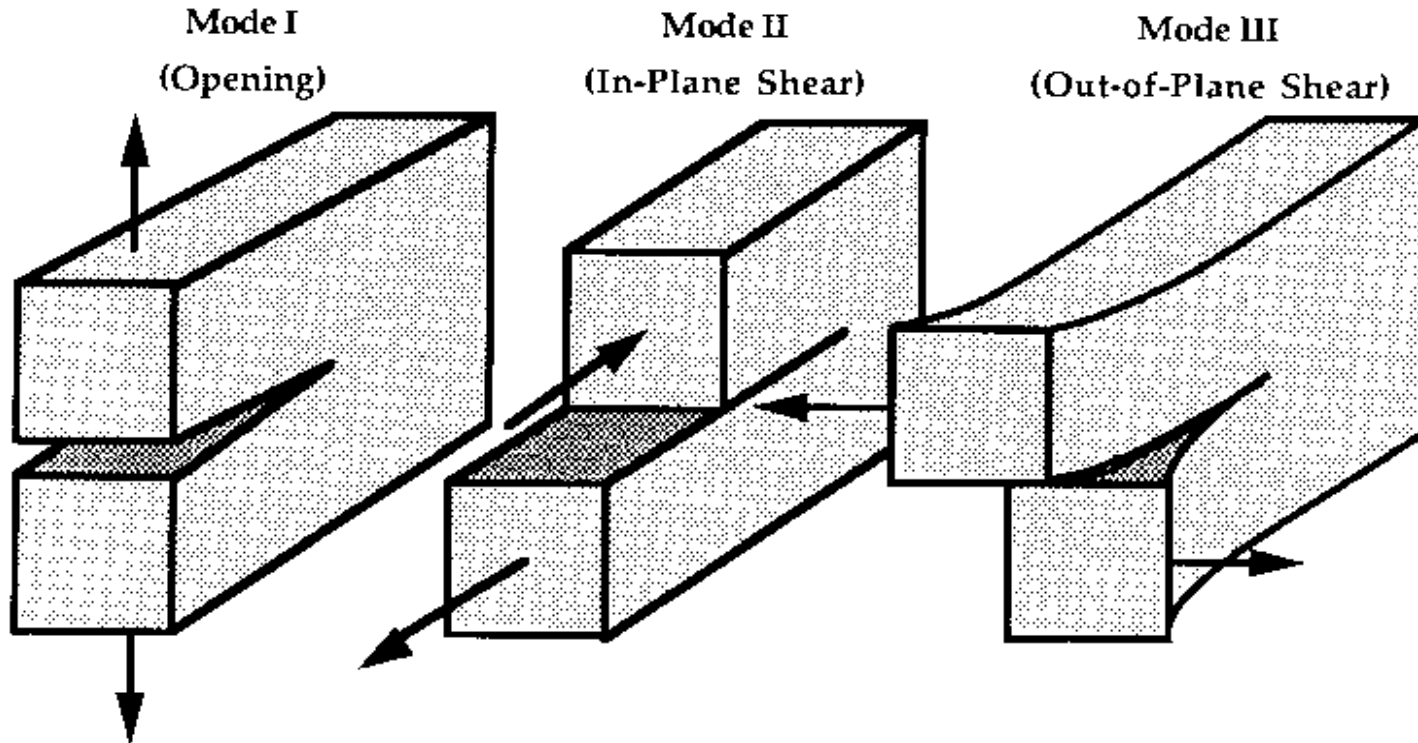
$$K_t = \frac{\sigma_A}{\sigma} = 1 + \frac{2a}{b}$$

$$K_t = 1 + 2\sqrt{\frac{a}{\rho}} \quad \text{where } \rho = \frac{b^2}{a}$$

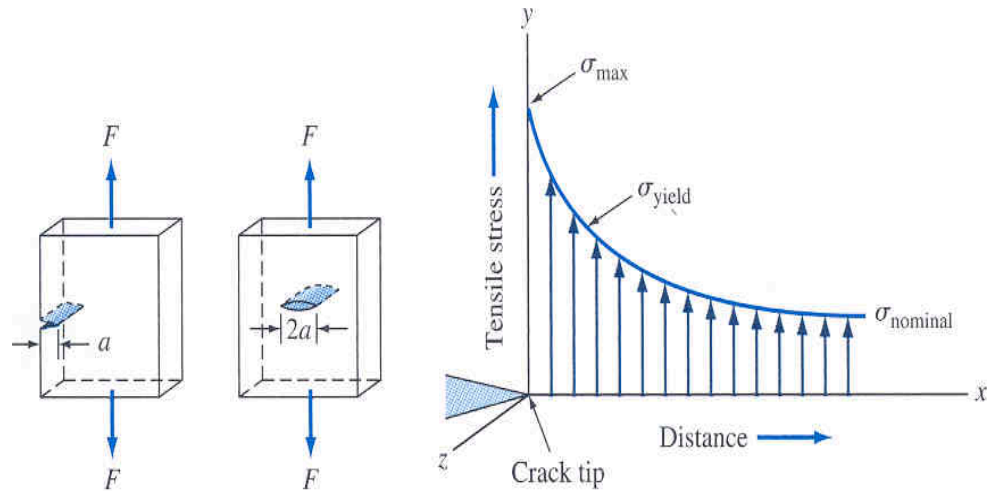
- ❑ Modeling a crack with a ellipse means $\rho \rightarrow 0$;
 $K_t \rightarrow \infty$, infinite stress
- ❑ K_t could not be used for crack problems



Loading Modes



The Stress Intensity Factor



$$K_I = Y \sigma \sqrt{\pi a}$$

K_I = Stress intensity factor.

σ = Applied stress.

a = edge crack length or
half the length of an
internal through crack

Y = geometric constant.

Unit of stress intensity factor K_I is $MPa\sqrt{m}$

Note: Y for center crack infinite plate is 1

Y for edge crack infinite plate is 1.12

Y for finite plates are given by polynomial function

The Stress Intensity Factor

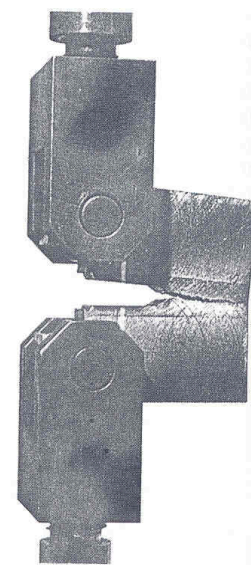
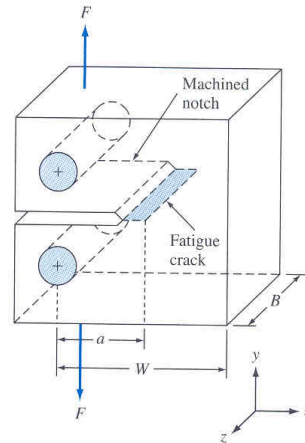
K_{Ic} = critical value of stress intensity factor (**Fracture toughness**)

$$K_{Ic} = Y \sigma_f \sqrt{\pi a}$$

Example:

Al 2024 T851	$26.2 \text{ MPa}\sqrt{\text{m}}$
4340 alloy steel	$60.4 \text{ MPa}\sqrt{\text{m}}$

- ❑ A **notch** is machined in a specimen of sufficient thickness B .
- ❑ $B \gg a$ for **plain strain** condition.
- ❑ $B = 2.5 \left(\frac{K_{IC}}{\sigma_{ys}} \right)^2$, where σ_{ys} is yield strength of the material
- ❑ Specimen is tensile tested.
- ❑ Higher the K_{Ic} value, more ductile the metal is.
- ❑ Used in design to find **allowable flaw size**.



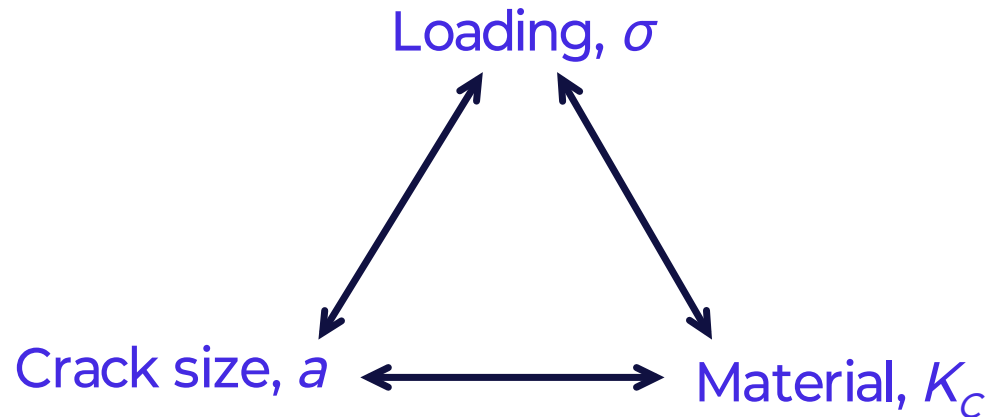


Importance of Fracture Mechanics

1. Selection of a Material
2. Design of a Component
3. Design of a manufacturing or Testing Method

Design based on Fracture Mechanics

$$K_c = Y \sigma_c \sqrt{\pi a_c}$$



Design based on Fracture Mechanics

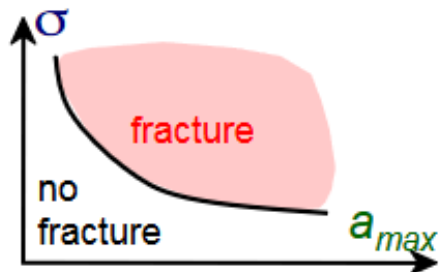
- Crack growth condition:

$$K \geq K_c = Y\sigma\sqrt{\pi a}$$

- Largest, most stressed cracks grow first!

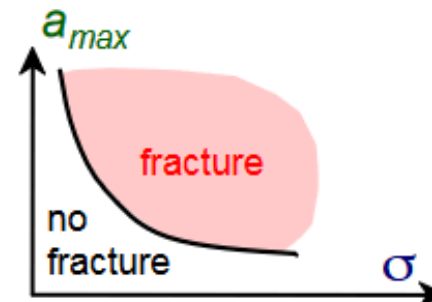
--Result 1: Max. flaw size dictates design stress.

$$\sigma_{design} < \frac{K_c}{Y\sqrt{\pi a_{max}}}$$



--Result 2: Design stress dictates max. flaw size.

$$a_{max} < \frac{1}{\pi} \left(\frac{K_c}{Y\sigma_{design}} \right)^2$$



Design based on Fracture Mechanics

Design Example: Aircraft Wing

Material has $K_c = 26 \text{ MPa}\sqrt{m}$

Two designs to consider...

Design A

--largest flaw is 9 mm

--failure occurs at stress = 112 MPa

Use....
$$\sigma_c = \frac{K_c}{Y \sqrt{\pi a_c}}$$

Design B

--use same material

--largest flaw is 4 mm

--failure stress= ?

Key point: Y and K_c are the same in both designs!

$$(\sigma_c \sqrt{\pi a_{\max}})_A = (\sigma_c \sqrt{\pi a_{\max}})_B$$

Answer $(\sigma_c)_B = 168 \text{ MPa}$

Reducing flaw size pays

off!



Fracture Toughness - K_C , K_{IC}

- ❑ Higher yield or ultimate strength generally produces a decrease in K_{IC} , and thus a greater susceptibility for catastrophic fracture!
- ❑ Low impurity materials provide better fracture toughness.
- ❑ K_{IC} can be very sensitive to metallurgical conditions such as grain orientation, chemistry, and microstructure.
- ❑ Fracture toughness K_C of metals is also dependent on temperature, strain rate, and corrosive environment.

Prob.1

A structural plate component of an engineering design must support 207 MPa in tension. If aluminum alloy 2024-T851 is used for this application, what is the largest internal flaw size that this material can support? (Use $Y = 1$).

■ Solution

$$K_{IC} = Y\sigma_f \sqrt{\pi a} \quad (7.2)$$

Using $Y = 1$ and $K_{IC} = 26.4 \text{ MPa}\sqrt{\text{m}}$ from Table 7.1,

$$a = \frac{1}{\pi} \left(\frac{K_{IC}}{\sigma_f} \right)^2 = \frac{1}{\pi} \left(\frac{26.4 \text{ MPa}\sqrt{\text{m}}}{207 \text{ MPa}} \right)^2 = 0.00518 \text{ m} = 5.18 \text{ mm}$$

Thus, the largest internal crack size that this plate can support is $2a$, or $(2)(5.18 \text{ mm}) = 10.36 \text{ mm}$.



References

1. William F. Smith, Javad Hashemi, 'Material Science & Engineering", In SI units Fourth edition, McGrawHill.
2. T. L. Anderson, Fracture Mechanics; Fundamentals and Applications.

Thank You